Transportation Research Part B 158 (2022) 164–186

Contents lists available at ScienceDirect

Transportation Research Part B

journal homepage: www.elsevier.com/locate/trb

Truthful double auction mechanisms for online freight platforms with transaction costs

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ARTICLE INFO

Keywords:
Logistics management
Transport procurement mechanism
Double auction
Mechanism design
Information asymmetry

ABSTRACT

In recent years, many online freight platforms have appeared to improve transport market efficiency. An important problem faced by such platforms is how to elicit the agents' information and to match their supplies and demands efficiently. To address the problem, we propose the modified Multi-unit Trading Reduction and Vickery–Clarke–Groves (MMTR-MVCG) mechanism for considering heterogeneous transaction costs. Under this mechanism, the shippers and carriers first decide their bids, and then the platform determines the trading agents, sets the transaction prices, and matches the shippers' demands and the carriers' supplies. We show that this mechanism is incentive-compatible, individually rational, budget-balanced, and asymptotically efficient. For the special case with homogeneous transaction costs, we first show that the MMTR-MVCG mechanism can be simplified. In addition, we design a new double auction mechanism for the case with asymmetric demand information and homogeneous transaction costs and show that this mechanism also is incentive- compatible, individually rational, budget-balanced, and asymptotically efficient. Moreover, we conduct numerical studies to assess the impacts of transaction costs and information asymmetry on the mechanisms and compare the performance of our mechanisms with those of two commonly used mechanisms. Our mechanisms can yield greater social welfare, more trading agents, and larger trading volumes than the other two mechanisms. We also discuss the impact of the subsidy strategy.

1. Introduction

It is well known that the transport market efficiency is very low worldwide. Eurostat (2018) reported that, at the total transport level, most member states in Europe fell in the range between 15% and 30% of empty truck journeys. Generally, a large number of small and private fleets and truckers occupy the supply market. For example, the American Trucking Associations reported that over 91.3% of carriers operated fewer than six trucks in the United States (ATA, 2020). Under this situation, carriers' transport ranges are small and they have little information on shippers. It is difficult for them to find return shipments. On the other hand, it is difficult for shippers to find proper carriers to provide transport services in the whole market. So they have to sign contracts with familiar fleets and truckers in different regions.

With the development of E-commerce (especially B2B), a large number of companies can purchase raw materials and sell products

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https://doi.org/10.1016/j.trb.2022.02.009
Received 20 August 2021; Received in revised form 10 February 2022; Accepted 14 February 2022
Available online 2 March 2022
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Table 1
The outline of this work

	Common knowledge of quantity	Private information on quantity
Heterogeneous transaction costs	Case (i) MMTR-MVCG mechanism	Case (iii)
Homogeneous transaction costs	Case (ii) Simplifying MMTR-MVCG mechanism	Case (iv) DA-AQI mechanism

nationally and even globally (Shim et al., 2000), which results in a great deal of unplanned and fragmented transport demands. For example, Dell Computer automates and streamlines the process of purchasing microprocessor chips and hard drives from suppliers through B2B e-commerce (Shim et al., 2000). These demands are difficult to consolidate into periodic and stable shipments under the traditional practice (Lafkihi et al., 2019). Under this situation, many online freight platforms (e.g., car-free carrier platforms) have emerged to integrate shipment and capacity information nationwide and to match shippers' demands with carriers' supplies in a short time. By using these platforms, shippers can reduce transport costs and carriers (individual drivers) can easily find return shipments. The online freight platforms have played an important role in integrating scattered transport capacity and reducing empty mileage since their emergence. In the first half of 2018, they integrated 644,000 vehicles and completed a total of 127 million tons of freight in China (MTPRC, 2018). According to the operational data of the Full Truck Alliance, the largest online freight platform in China, the monthly mileage of drivers using the platform increased from 9,000 km to 135,000 km, and the vehicle no-load rate decreased from 37% to 32%, from its inception to 2018 (SIC, 2019).

The operations of an online freight platform are as follows: The platform first announces its rules (bid rule, allocation rule, payment rule, etc.). Then shippers bid their demands (lane, volume, and willingness payment) and carriers bid their supplies (lane, capacity, and cost). After receiving the bids, the platform decides the transaction prices, allocates the winning shippers' demands to the winning carriers, and decides the winners' payments. The online freight platform is a bilateral trade market for transport service procurement (TSP).

For the TSP problem, auction mechanisms are commonly used in practice and addressed by researchers, accounting for 83% of the papers in Lafkihi et al.'s (2019) review on TSP. A majority of these works study the one-sided auction through which a single shipper procures his transport service from a group of carriers. When the one-sided auction is used to select carriers, each shipper has to hold the auction independently to choose his carriers. However, one-sided value maximization is not equivalent to social welfare maximization when there are many shippers and carriers in a large transport market (Huang and Xu, 2013). Also, to win a shipment, a carrier may bid repeatedly in different auctions held by different shippers and contemplate the possible outcomes in each auction, which imposes a significant computational burden (Chu, 2009). Double auctions are more time-efficient and practically attractive than one-sided auctions in such a transport market (Xu et al., 2017).

Huang and Xu (2013) first adopted the double auction theory to study the TSP mechanism design in which both shippers and carriers submit their demands and supplies. It determines the optimal transport and exchange relationships based on the agents' bids to maximize social welfare. Subsequently, Xu and Huang (2013), Xu et al. (2017), and Liang et al. (2020) studied similar problems. These works assume that the demand and supply quantities are *common knowledge* and transaction costs *are zero*. In reality, an agent does not know the other agents' demands or supplies, so the demand and supply information is generally private. The agents have an incentive to lie to gain more profits. For example, a shipper may exaggerate its transport demands to secure more transport services. Similarly, a shipper may under-report his demand to reduce the market demand and drive down the market price. Moreover, transport trading often incurs transaction costs. Xu et al. (2015) pointed out that transaction costs generally include costs associated with the potential risks during transport, service quality, the locations of shippers and carriers, and shipper–carrier relationship, etc. However, transaction costs are often overlooked in the design of the auction mechanism. An auction mechanism without considering transaction costs can affect the final allocation and reduce the realized social welfare. For example, there are two shippers, two carriers, and an online freight platform in the transport market. Shipper 1 and shipper 2 want to purchase one unit of transport service, for $600 and $400, respectively. Carrier 1 and carrier 2 want to sell one unit of transport service for $400 and $450, respectively. Because the shippers and carriers are in different locations, carrier 1 and carrier 2 will incur a transaction cost of $100 and $30, respectively, to trade with any shipper. The platform makes decisions to maximize social welfare. Without considering transaction costs, only shipper 1 procures one unit of transport service from carrier 1, yielding overall social welfare of $200. However, the transaction incurs a transaction cost of $100 and the final realized social welfare is $100. If transaction costs are taken into account in the decision, then shipper 1 procures one unit of transport service from carrier 2, where the realized social welfare is $120, which is greater than $100. In this study, we design truthful double auction mechanisms under the assumption that transaction costs are non-negative, and the shippers' demand and carriers' supply quantity information are asymmetric. For transaction costs, there are two cases, namely heterogeneous and homogeneous. For demand/supply quantity information, there are also two cases, namely common knowledge and private information. Then there are four cases as shown in Table 1.

We first propose the MMTR-MVCG mechanism for Case (i) based on the ideas of the multi-unit trading reduction (MTR) mechanism and the Vickery–Clarke–Groves (VCG) mechanism. We show that the mechanism is incentive-compatible, individually rational, budget-balanced, and asymptotically efficient. For Case (ii), we show that the MMTR-MVCG mechanism can be simplified to reduce the computational burden. For Case (iv), we propose a new double auction with asymmetric quantity information, i.e., the DA-AQI mechanism, and show that it is incentive-compatible, individually rational, budget-balanced, and asymptotically efficient. It is very difficult to design a truthful double auction mechanism for Case (iii), so we leave it for future study. Then we conduct numerical studies to assess the impacts of transaction costs and demand information asymmetry on the mechanisms and compare the performance of our mechanisms with those of the mechanism used by the FOR-U Trucking and the well-known MTR mechanism. We further consider a

subsidy strategy and discuss its impacts on the performance of our mechanism.

Compared with the existing literature, this paper makes the following contributions:

(1) We first design a double auction mechanism, i.e., the MMTR-MVCG mechanism, which is incentive-compatible, individually rational, budget-balanced, and asymptotically efficient, for the TSP with transaction costs between shippers and carriers;
(2) For the case with homogeneous transaction costs (e.g., the transaction costs are only related to the shippers) and where the supply/demand quantity information is common knowledge, we simplify the MMTR-MVCG mechanism and reduce its computational burden;
(3) For the case with homogeneous transaction costs and private demand quantity information, we design a novel mechanism, i.e., the DA-AQI mechanism, which is incentive-compatible, individually rational, budget-balanced, and asymptotically efficient;
(4) We discuss the impacts of subsidies on agents' bidding and the performance of the double auction mechanism. We find that a subsidy can attract more shippers and carriers and improve the trading efficiency in the long run;
(5) We conduct numerical experiments to compare the performance of our mechanisms with those of two commonly used mechanisms. The comparison shows that our mechanisms can yield greater social welfare, more trading agents, and larger trading volumes than the other two mechanisms.

The rest of the paper is organized as follows: In Section 2 we review the relevant literature to identify the research gap and position our study. In Section 3 we discuss the bilateral trading environment in the transport market. We design a double auction mechanism for the market with heterogeneous transaction costs in Section 4. In Section 5 we consider the special case with homogeneous transaction costs. In Section 6 we present the numerical studies and generate managerial insights from the analytical findings. In Section 7 we discuss the impact of subsidy strategy. Finally, in Section 8, we conclude the paper and suggest topics for future study. We present the proofs in the Appendix.

2. Literature review

In this paper, we study the double auction mechanism design for the online freight platform with transaction costs. There are three streams of literature related to this work, namely the reverse auction for TSP, the double auction for TSP, and the mechanism design incorporating transaction costs.

The first stream of literature related to this study is on reverse auctions in the transport market. There is a growing body of literature studying the reverse auction for the TSP problem (Xu and Huang, 2014). Ledyard et al. (2002) studied the application of the combination auction for the TSP problem of Sears Logistics Services, where 3PLs could bid on combinations of the transport routes. Sheffi (2004), Day and Raghavan (2009), and Remli and Rekik (2013) investigated the problem of setting the winning bid price in the reverse auction for TSP. Huang and Xu (2013) addressed the problem of the reverse auction for multi-unit TSP by proposing an O-VCG mechanism. Xu et al. (2015) investigated the intermodal transport auction for the B2B e-commerce logistics problem, in which each 3PL submits a combinatorial bid. Distinct from the above price-only reverse auction, some studies consider the multi-attribute reverse auction for TSP. Guo et al. (2006) studied winner determination considering transport service quality. Huang et al. (2016) studied the same problem with the multi-attribute reverse auction considering a risk-averse buyer. Hu et al. (2016) proposed a two-objective integer planning model for TSP considering price and transport time. Xu and Huang (2017) studied the matching problem between B2B e-commerce's online orders and 3PL providers considering the price and some non-monetary attributes. Zhang et al. (2019) proposed a multi-attribute TSP mechanism for the case where the carriers have multiple units of transport capacity.

The second stream of literature concerns the double auction, where multiple buyers and multiple sellers participate in the auction held by a market broker. McAfee (1992) first proposed the trade reduction double auction mechanism for the case where multiple buyers and sellers exchange a unit of goods. Babaioff and Walsh (2005) proposed a known single-minded trade reduction mechanism for the case where each buyer wants to purchase a bundle of commodities and each seller offers one unit of a single commodity under the known single-minded assumption and the single output restriction. Chu and Shen (2006) designed the agent competition double-auction (AC-DA) mechanism for a bilateral trading environment, where each buyer and each seller trades one unit of a single commodity. Chu and Shen (2008) used the multi-stage design approach to design a linear planning buyer competition (BC-LP) mechanism and a modified buyer competition (MBC) mechanism, in which the transaction prices are determined jointly with the shadow price and the VCG price. These two mechanisms can be applied in the bilateral trading environment under the known single-minded assumption and the single output restriction. Chu (2009) designed a double auction mechanism by introducing the "virtual buyers" with unlimited budgets for the case where each seller could offer multiple units of a single commodity. Cheng et al. (2016) extended the mechanisms of Chu and Shen (2008) and Chu (2009) to the case with multi-attribute perishable products.

Some works study the double auction mechanism for TSP. Huang and Xu (2013) proposed the multi-unit trade reduction (MTR), MTR-buyer augment (MTR-BA), and MTR-seller augment (MTR-SA) double auction mechanisms to guide the logistics e-marketplace to clear the market. Xu and Huang (2013) devised a periodic sealed double auction (PSDA) mechanism to clear markets with stochastic but balanced supply and demand for dynamic transport services. Xu et al. (2017) proposed truthful budget-balanced bundle double auction mechanisms for carrier collaboration over a bilateral exchange transport network. Sun et al. (2019) designed three double auction mechanisms for the intermodal TSP problem in the context of the Belt and Road Initiative. Liang et al. (2020) designed a truthful double auction mechanism with quantity discounts and cooperative procurement for TSP. The above studies assume that the demand and supply quantities are common knowledge and transaction costs are zero. We consider positive transaction costs and asymmetric quantity information.

Table 2
The relation of this study to the existing literature

Literature	Double auction	Asymmetric quantity information	Transaction costs	Heterogeneous transaction costs
McAfee (1992)	√	×	×	×
Noussair et al. (1998)	√	×	√	×
Babaioff and Walsh (2005)	√	√	×	×
Chu and Shen (2006)	√	×	√	√
Chu and Shen (2008)	√	×	√	√
Chu (2009)	√	×	×	×
Huang and Xu (2013)	√	×	×	×
Xu and Huang (2013)	√	×	×	×
Xu et al. (2015)	×	×	√	×
Cheng et al. (2016)	√	×	×	×
Lloret-Batlle et al. (2017)	√	×	√	√
Masoud et al. (2017)	×	×	√	√
Xu et al. (2017)	√	×	×	×
Zhang et al. (2019)	×	√	×	×
Sun et al. (2019)	√	×	√	×
Liang et al. (2020)	√	×	×	×
Current paper (MMTR-MVCG)	√	×	√	√
Current paper (DA-AQI)	√	√	√	×

The third stream of research focuses on auction mechanism design considering transaction costs. Noussair et al. (1998) showed that the presence of transaction costs leads to lower quantity exchanged and market efficiency than in competitive equilibrium. Chu and Shen (2006) designed a mechanism for a two-sided exchange environment with pair-related transaction costs, in which each agent wants to buy (sell) one unit of the same item. Chu and Shen (2008) studied two double auction mechanisms with transaction costs and showed that they are asymptotically efficient. They considered heterogeneous transaction costs in an exchange environment where each buyer wants to procure a bundle of goods and each seller provides one unit of a single item. Xu et al. (2015) proposed one-sided auction mechanisms to minimize the total logistics chain costs for outsourcing the logistics services of B2B platforms considering transaction costs. Sun et al. (2019) designed double auction mechanisms considering the homogeneous transaction costs related to transport lanes. Each shipper wants to purchase a bundle of transport services. The above works assume that the supply and demand quantities are common knowledge.

The mechanisms for passenger transport consider the initial costs of drivers to acquire passengers, which are similar to the transaction costs. Wang et al. (2016), He et al. (2018), and Zhang and Nie (2021) studied the pricing strategies of taxi-hailing platforms (pooling e-hail platforms) for information or transport service. Different from them, we focus on mechanism design to match supply and demand efficiently. Tafreshian and Masoud (2020) designed a joint matching and payment mechanism for the peer-to-peer ride-sharing problem assuming that the agents' information is common knowledge. Lloret-Batlle et al. (2017) proposed a double VCG mechanism to match the drivers and passengers, which is not budget-balanced. Masoud et al. (2017) designed a posed-price mechanism to clear the market for peer-to-peer ride-sharing and showed that the mechanism is incentive-compatible, budget--balanced, and individually rational but not asymptotically efficient. Here we use other ideas to design double auction mechanisms and show that our mechanism is incentive-compatible, budget-balanced, individually rational, and asymptotically efficient. In addition, we further consider asymmetric quantity information.

Different from the above works, we study the problem of multi-unit double auction mechanism design with transaction costs and private information on the demand quantities. By introducing an *ex-post* penalty for the carriers, Zhang et al. (2019) proposed a truthful multi-attribute reverse auction mechanism for TSP with carriers having multi-unit supplies, which are private information. But they only considered one buyer and a one-sided auction. Babaioff and Walsh (2005) proposed a truthful unknown single-minded trade reduction mechanism (USM-TR) for the case where each buyer wants to purchase a bundle of commodities that is private information and each seller offers one unit of a single commodity. However, they showed that the USM-TR mechanism fails incentive compatibility without the single-minded restriction where all the agents submit their "dividable" multi-unit bids. Huang and Xu (2013) also pointed out that it is very difficult to design a truthful multi-unit auction mechanism when the demand and supply quantities are private information. It is not clear how a mechanism decides the allocation and determines the transaction prices to guarantee incentive compatibility with budget balance, without reliable information on demand and supply quantities.

Table 2 summarizes the relation of this paper to the existing literature.

3. Problem settings

Consider a set of shippers I and a set of carriers J that want to purchase and sell transport service on an online freight platform (we refer to a shipper as "he" and a carrier as "she", and call them agents in general). Suppose that the shipments are on a lane. One truckload transport service is regarded as "one-unit service", which is the minimum trade unit with the full truckload (FTL) constraint. Shipper $i(i \in I)$ wants to purchase X_i units of service while carrier $j(j \in J)$ wants to sell Y_j units of service. We use the term "bid" to denote an agent's declaration. A shipper (carrier) bids for the transport demand (supply) to the platform in the form of {bid price ($\widehat{s}_i$), bid demand (X_i)} ({bid cost ($\widehat{c}_j$), bid supply (Y_j)}). We assume that the bids are "dividable", i.e., each shipper (carrier) can split his (her) bid

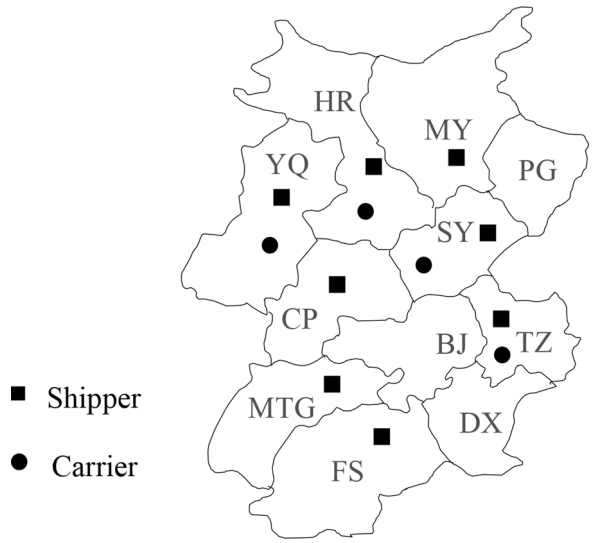


Fig. 1. An example with transaction costs.

volume, but not less than one truckload (Nisan, 2000). Suppose that shipper i's willingness-to-pay for one-unit service s_i and carrier j's cost of one-unit service c_j are private information. Assume the demand quantity X_i and supply quantity Y_j are common knowledge in this section and the next section (In Subsection 5.2, we shall relax this assumption and consider the case that demand quantity X_i is private information). Assume that there is no collusion among the agents in the market. Similar assumptions can be found in many studies on double auction mechanisms for the transportation market, such as Huang and Xu (2013), Xu and Huang (2013), and Liang et al. (2020).

Suppose that each agent submits its bid to maximize its utility, which is quasi-linear. That is, an agent's utility is the difference between the real valuation and the money transferred. And an agent's utility is zero if it is not involved in the final trade. The platform's payoff is the difference between the total payment from the shippers and the total transfer to the carriers. Social welfare is defined as the sum of the agents' utilities and the platform's payoff.

Transactions in this transport market may incur transaction costs. Let the transaction cost incurred when shipper i purchases transport service from carrier j be d_{ij}. To facilitate expression and measurement, the transaction costs in this paper concern the costs associated with the geographic locations of the carriers and shippers (see Fig. 1). They can also be extended to include the costs associated with service quality, lead time, etc. The transaction costs are paid by the shippers and then transferred to the carriers via the platform, which are common knowledge.

We list the notation used throughout the paper as follows:

I/J: Set of shippers/carriers (indexed by i/j);
$\bar{I}/\bar{J}$: Set of remaining shippers/carriers after the first allocation;
$\widetilde{I}/\widetilde{J}$: Set of final trading shippers/carriers;
X_i/Y_j: Transport service shipper i/carrier j wants to purchase/sell;
$\widehat{X}_i/\widehat{Y}_j$: Transport service volume bid by shipper i/carrier j;
s_i/c_j: Real valuation/cost per unit service of shipper i/carrier j;
$\widehat{s}_i/\widehat{c}_j$: Bid price per unit item bid by shipper i/carrier j;
α_I/α_J: Subsidies paid by the platform to shipper i/carrier j;
p^+/p^-: Shipper/ carrier threshold price;
$\widetilde{P}_i/\widetilde{P}_j$: Total transaction payment (revenue) for shipper i/carrier j;
p_i^{VCG}/p_j^{VCG}: VCG payment (revenue) for shipper i/carrier j;
$(\widetilde{P}_i)^S/(\widetilde{P}_j)^S$: Total transaction payment/revenue for shipper i/carrier$j$ with subsidies;
$(\widetilde{I})^S/(\widetilde{J})^S$: Set of trading shippers/trading carriers with subsidies;
$V(I, J)$: Social welfare corresponding to the shipper set I and the carrier set J;
$V(I, J)^{-i}/V(I, J)^{-j}$: Social welfare after removal of shipper i/ carrier j;
d_{ij}: Transaction costs when shipper i purchases transport service from carrier j;
Q: Total demand of remaining shippers;
C: Total supply of remaining carriers;
q_{ij}: Quantity of transport service purchased by shipper i from carrier j;
$\bar{q}_{ij}$: Quantity of transport service purchased by remaining shipper i from remaining carrier j;

Fig. 2 shows the transaction process of the online freight platform under this study.

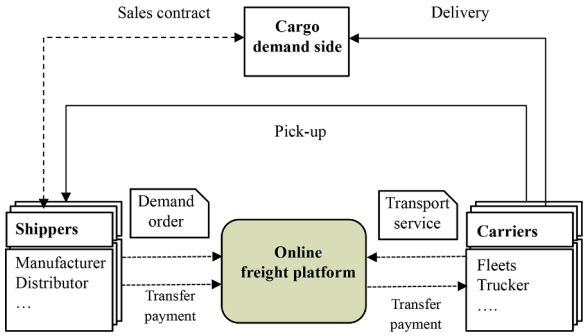


Fig. 2. Transaction process of the online freight platform.

The sequence of events is as follows:

(1) The platform announces the bid rule, winner determination rule, payment rule, and matching rule;
(2) The shippers submit their bids based on their valuation and demand to the online freight platform;
(3) The carriers submit their bids based on their transport capacity and costs to the online freight platform;
(4) The platform determines the final trading shippers and carriers according to the winner determination rule, sets the transaction prices according to the payment rule, and matches the shipments according to the matching rule;
(5) The final trading carriers and shippers sign contracts;
(6) The contracts are implemented and the payments are transferred.

The problem faced by the platform is to design a double auction mechanism (specifying the winner determination rule, payment rule, and matching rule) to induce the shippers and carriers to participate willingly (individual rationality), to tell their information truthfully (incentive compatibility), to match the supplies and demands efficiently, and to make a profit (budget balance).

4. Double auction mechanism with heterogeneous transaction costs

In this section, we consider the mechanism design for the auction with heterogeneous transaction costs. In this case, the transaction costs depend on the carriers and shippers. Then the agents are heterogeneous although they submit the same bid.

4.1. Finding the maximum social welfare

In this subsection, we determine the maximum social welfare if all the agents bid truthfully. The maximal social welfare can be found by solving the following integer programming problem.

$$\boldsymbol{P1}(I,J):\max V(I,J)=\sum_{i=1}^{m}\sum_{j=1}^{n}q_{ij}\left(s_i-c_j-d_{ij}\right) \tag{1}$$

$$s.t.\sum_{j=1}^{n}q_{ij}\le X_i, i\in I \tag{2}$$

$$\sum_{i=1}^{m}q_{ij}\le Y_j, j\in J \tag{3}$$

$$q_{ij}\ge 0, q_{ij}\in\mathbb{Z}, \forall i,j. \tag{4}$$

where constraints (2) and (3) mean that the shippers cannot procure more than their needs and the carriers cannot sell more than their capacities, respectively.

To solve the problem ***P1****(I, J)*, we consider its linear relaxation as follows:

$$\boldsymbol{LR1}(I,J):\max V(I,J)=\sum_{i=1}^{m}\sum_{j=1}^{n}q_{ij}\left(s_i-c_j-d_{ij}\right) \tag{5}$$

$$s.t.\sum_{j=1}^{n}q_{ij}\le X_i, i\in I \tag{6}$$

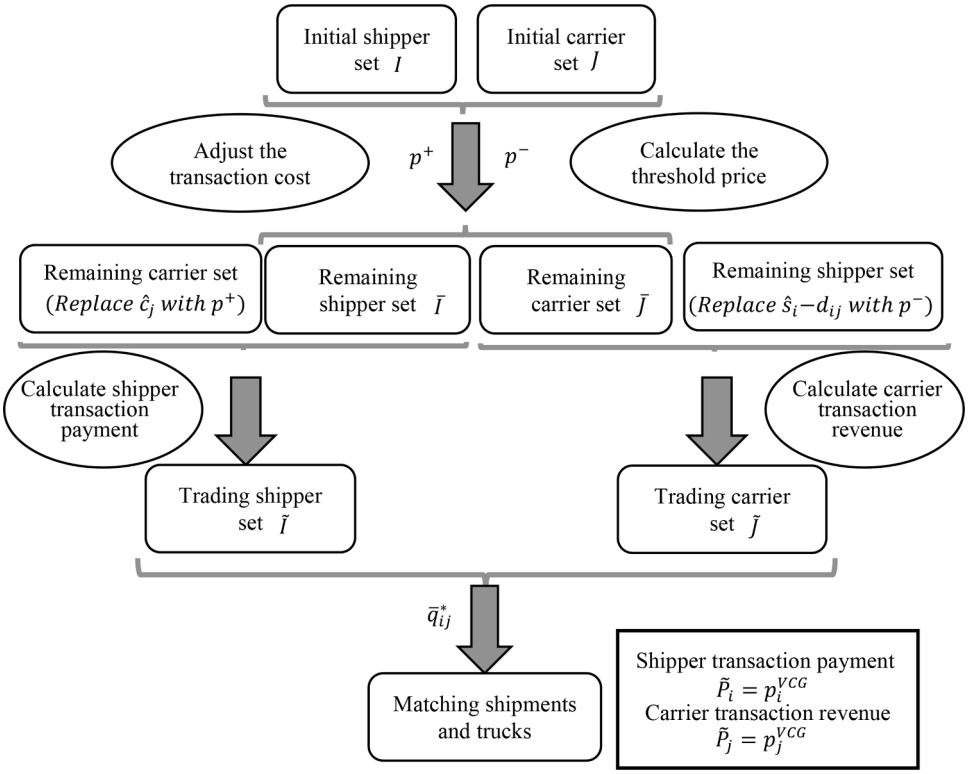


Fig. 3. The MMTR-MVCG mechanism.

$$\sum_{i=1}^{m} q_{ij} \le Y_j, j \in J \tag{7}$$

$$q_{ij} \ge 0, \forall i, j. \tag{8}$$

Theorem 1 shows that the solution of the linear programming problem ***LR1***(I, J) is a solution of the integer programming problem ***P1***(I, J).

Theorem 1. *The optimal solution of* ***LR1***(I, J) *is also an optimal solution of* ***P1***(I, J).

Proof. Please see Appendix A.1.

Note that there may be multiple optimal solutions to ***LR1***(I, J). We use the perturbation technique proposed by Babaioff and Walsh (2005) to select a single optimal solution of ***LR1***(I, J). The perturbation factor can be interpreted as an agent's trading history, credit, and other relevant factors in practice. Agents with the same bid price can be prioritized according to these factors. We take the unique optimal solution of ***LR1***(I, J) with perturbation factors as an efficient allocation of the initial bids.

Theorem 1 states that the maximal social welfare can be obtained by solving the linear programming problem ***LR1***(I, J). This means that the online freight platform can easily obtain an efficient transaction that maximizes social welfare if all of the agents bid truthfully. Note that the shippers' demands and the carriers' supplies are private information, so we need to design a direct truthful double auction mechanism to induce the agents to bid truthfully from the revelation principle (Myerson, 1981).

4.2. The MMTR-MVCG double auction mechanism

In this subsection, we state the MMTR-MVCG double auction mechanism in detail. The mechanism has two steps: First, we use the modified MTR mechanism (the MTR mechanism was developed by Huang and Xu (2013) for the case with zero transaction costs and common knowledge on quantities) to set the threshold prices for agents. The shippers with bid prices not higher than the shipper threshold price and the carriers with bid prices not lower than the carrier threshold price are removed. In the second stage, we propose a modified VCG mechanism to set the transaction prices. Then we make an efficient allocation among the remaining agents to determine the final trading agents and shipment allocation. We call this mechanism the MMTR-MVCG mechanism. To state the mechanism conveniently, we call the difference of shipper i's bid price and the transaction cost, $\hat{s}_i - d_{ij} (i \in I, j \in J)$, as his total bid price for carrier j. Note that shipper i's total bid prices $\hat{s}_i - d_{ij} (i \in I, j \in J)$ are different for different carriers.

Fig. 3 shows the procedure of the MMTR-MVCG mechanism.

First, we decide the shipper threshold price p^+ and the carrier threshold price p^- that are used to remove some agents which are inefficient for the trade. Different from the traditional MTR mechanism (Huang and Xu, 2013), we directly use the solution of the linear programming problem ***LR1***(I, J) to make the first allocation and decide p^+ and p^-. Given heterogeneous transaction costs, the lowest shipper bid price and the highest carrier bid price in the first allocation may or may not lie at the equilibrium point of trading. Then this lowest shipper bid price may be less than the highest carrier bid price in the efficient allocation. Because of this, we set the shipper threshold price as the maximum of the two prices and the carrier threshold price as the minimum. Thus, the shipper threshold price p^+ and the carrier threshold price p^- can be determined as follows, where $q_{ij}^*(i \in I.j \in J)$ is the optimal solution of ***LR1***(I, J) and $x \vee y = \max\{x, y\}$, $x \wedge y = \min\{x, y\}$:

$$p^+ = \min\{\widehat{s}_i - d_{ij} | q_{ij}^* > 0, i \in I, j \in J\} \vee \max\{\widehat{c}_j | \sum_{i \in I} q_{ij}^* > 0, j \in J\}, \tag{9}$$

$$p^- = \min\{\widehat{s}_i - d_{ij} | q_{ij}^* > 0, i \in I, j \in J\} \wedge \max\{\widehat{c}_j | \sum_{i \in I} q_{ij}^* > 0, j \in J\}. \tag{10}$$

Note that the total bid prices $\widehat{s}_i - d_{ij}$ of shipper i are different for different carriers, and the rule that a shipper is removed from the trade if any one of his total bid prices is less than p^+ will result in a loss of market efficiency. Because of this, for shipper i, we only remove the trading relationship between shipper i and carrier j when the total bid price $\widehat{s}_i - d_{ij}$ of shipper i is not higher than p^+. If the trading relationship between shipper i and carrier j is removed, we set the corresponding transaction cost as infinity, i.e., $d_{ij} = \infty$. Otherwise, the transaction cost remains unchanged. For the carriers, we remove those whose corresponding bid prices are not lower than the carrier threshold price. Thus, the set of the remaining shippers, $\overline{I}$, and the set of the remaining carriers, $\overline{J}$, are determined as follows:

$$\overline{I} = \{i | \exists d_{ij} \neq \infty, i \in I, j \in J\}, \tag{11}$$

$$\overline{J} = \{j | \widehat{s}_j < p^- \text{and} \exists d_{ij} \neq \infty, i \in I, j \in J\}. \tag{12}$$

In the second stage, the sets of final trading agents are determined as $\widetilde{I} = \{i | \sum_{j \in \overline{J}} \overline{q}_{ij}^* > 0, i \in \overline{I}\}$ and $\widetilde{J} = \{j | \sum_{i \in \overline{I}} \overline{q}_{ij}^* > 0, j \in \overline{J}\}$, where $\overline{q}_{ij}^*$ is the optimal solution of ***LR1***$(\overline{I}, \overline{J})$. Trading shipper $i(i \in \widetilde{I})$purchases $\overline{q}_{ij}^*(i \in \widetilde{I}, j \in \widetilde{J})$ units of transport service from trading carrier$j(j \in \widetilde{J})$.

In terms of the payment rule, we adopt the VCG mechanism to set the shippers' and carriers' transaction payment to induce them to bid truthfully. Shipper i's VCG payment and carrier j's VCG revenue can be determined as follows:

$$p_i^{VCG} = \widehat{s}_i \sum_{j \in J} q_{ij}^* - \left(V(I, J) - V(I, J)^{-i}\right), i \in I, \tag{13}$$

$$p_j^{VCG} = \sum_{i \in I} q_{ij}^* \left(\widehat{c}_j + d_{ij}\right) + \left(V(I, J) - V(I, J)^{-j}\right), j \in J. \tag{14}$$

Note that the VCG mechanism is not budget-balanced in the bilateral trading environments. To overcome this problem, we use the shipper threshold price p^+ to replace the bid price $\widehat{c}_j$ of remaining carrier $j(j \in \overline{J})$ and set the total transaction payment $\widetilde{P}_i$ of trading shipper $i(i \in \widetilde{I})$ to the corresponding VCG payment p_i^{VCG} as follows:

$$\widetilde{P}_i = p_i^{VCG} = \widehat{s}_i \sum_{j \in \overline{J}} \overline{q}_{ij}^* - V\left(\overline{I}, \overline{J} | \widehat{c}_j = p^+\right) + V\left(\overline{I}, \overline{J} | \widehat{c}_j = p^+\right)^{-i}, i \in \widetilde{I}. \tag{15}$$

Similarly, we use the carrier threshold price p^- to replace the total bid price $\widehat{s}_i - d_{ij}$ of remaining shipper $i(i \in \overline{I})$. Then we set the total transaction revenue $\widetilde{P}_j$ of trading carrier $j(j \in \widetilde{J})$ to be the corresponding VCG revenue p_j^{VCG} as follows:

$$\widetilde{P}_j = p_j^{VCG} = \sum_{i \in \overline{I}} \overline{q}_{ij}^* \left(\widehat{c}_j + d_{ij}\right) + V\left(\overline{I}, \overline{J} | \widehat{s}_i - d_{ij} = p^-, d_{ij} \neq \infty\right) - V\left(\overline{I}, \overline{J} | \widehat{s}_i - d_{ij} = p^-, d_{ij} \neq \infty\right)^{-j}, j \in \widetilde{J} \tag{16}$$

Now we state the procedure of the MMTR-MVCG mechanism in detail as follows:

Step 1: Shipper i submits a sealed bid $\{\widehat{s}_i, X_i\}$ and carrier j submits a sealed bid $\{\widehat{c}_j, Y_j\}$. The perturbation technique is used to make the total bid price $\widehat{s}_i - d_{ij}$ of shipper i and the bid price $\widehat{c}_j$ of carrier j different.

Step 2: Find the optimal solution q_{ij}^* of ***LR1***(I, J) and calculate the shipper threshold price p^+ and the carrier threshold price p^- using (9) and (10), respectively.

Step 3: For shipper $i(i \in I)$ and carrier $j(j \in J)$, if $s_i - d_{ij} \leq p^+$, we set d_{ij} to be ∞.

Step 4: Determine the remaining shipper set $\overline{I} = \{i | \exists d_{ij} \neq \infty, i \in I, j \in J\}$ and the remaining carrier set $\overline{J} = \{j | \widehat{s}_j < p^- \text{and} \exists d_{ij} \neq \infty, i \in I, j \in J\}$.

Winner determination rule

Step 5: Find the optimal solution $\overline{q}_{ij}^{*}$ of $\boldsymbol{LR1}(\overline{I},\overline{J})$ to determine the trading shipper set $\widetilde{I}=\{i\Big|\sum_{j\in\overline{J}}\overline{q}_{ij}^{*}>0,i\in\overline{I}\}$ and the trading carrier set $\widetilde{J}=\{j\Big|\sum_{i\in\overline{I}}\overline{q}_{ij}^{*}>0,j\in\overline{J}\}$.

Payment rule for trading shippers

Step 6: Replace the bid price $\widehat{c}_j$ of remaining carrier $j(j\in\overline{J})$ with the shipper threshold price p^{+} and decide trading shipper i's $(i\in\widetilde{I})$ transaction payment $\widetilde{P}_i$ using (15).

Payment rule for trading carriers

Step 7: Replace the total bid price $\widehat{s}_i-d_{ij}$ of remaining shipper $i(i\in\overline{I})$ with the carrier threshold price p^{-} and decide trading carrier j's $(j\in\widetilde{J})$ transaction revenue $\widetilde{P}_j$ using (16).

Matching rule

Step 8: Trading shipper $i(i\in\widetilde{I})$ purchases $\overline{q}_{ij}^{*}$ units of transport service from trading carrier $j(j\in\widetilde{J})$.

Sun et al. (2019) proposed mechanisms for the case where the demands are indivisible and transaction costs are only related to lanes. They used the trade reduction design approach that removes the least profitable trade subset from the efficient allocation. The bids in the removed trade determine the transaction price. Their mechanism is not incentive-compatible for carriers with multi-unit supplies. Here, we use the modified VCG mechanism to determine the carriers' and shippers' payments, so our mechanism is incentive-compatible for both carriers and suppliers. Huang and Xu (2013) also used the trade reduction approach to design the MTR mechanism for the case with zero transaction costs. In their mechanism, shippers (carriers) with large demand (supply) are assigned first. But the agents with larger quantities do not necessarily have high valuations or low costs. So their mechanism may lose some social welfare. To overcome such shortcoming, we use the modified VCG mechanism to match supplies and demands among the remaining agents efficiently. The BC-LP mechanism and MBC mechanism in Chu and Shen (2008) begin with a pricing decision for the buyers, which is used to remove some buyers, and then make the allocation and pricing decisions for the sellers using the shadow prices or VCG prices. These two mechanisms are not budget-balanced if shippers have multi-unit supplies. In our mechanism, we first use the solution of the linear programming $\boldsymbol{LR1}(I, J)$ to remove both shippers and carriers that are not involved in the trade. We then use a modified VCG mechanism to set the transaction prices and make efficient resource allocation among the remaining agents. The modified VCG mechanism that we propose can incentivize the agents to bid truthfully and generate budget surplus for the auctioneer for the case with multi-unit demands and supplies.

4.3. Properties

In this subsection, we show that the MMTR-MVCG mechanism is incentive-compatible, individually rational, budget-balanced, and asymptotically efficient.

Theorem 2. *The MMTR-MVCG mechanism is incentive-compatible.*

Proof. Please see Appendix A.2.

Theorem 2 states that the shippers (carriers) will bid their real valuations (costs) truthfully in equilibrium. So they do not manipulate their bid prices.

Theorem 3. *The MMTR-MVCG mechanism is (ex post) individually rational and (ex post weakly) budget-balanced.*

Proof. Please see Appendix A.3.

Theorem 3 shows that the shippers and carriers can get non-negative profits under the MMTR-MVCG mechanism, so potential shippers and carriers are willing to participate in the transaction. Moreover, the MMTR-MVCG mechanism can ensure that the platform gets a non-negative profit.

To prove that the MMTR-MVCG mechanism is asymptotically efficient, we assume that the shippers' valuations independently and identically follow the same distribution $F(s_i)_{i\in I}$, and that the carriers' costs independently and identically follow the same distribution $G(c_j)_{j\in J}$. Also, the transaction cost d_{ij} is proportional to the distance between shipper i and shipper j, who are independently and identically distributed according to some continuous distribution U on some compact domain H, and the distance function is continuous with respect to the agents' locations.

Theorem 4. *Supposing that the valuation and cost distributions are bounded and the transaction costs are continuous, the MMTR-MMVCG mechanism is asymptotically efficient.*

Proof. Please see Appendix A.4.

Theorems 2–4 generalize the results in Chu and Shen (2008) to the case where the agents' bids are multi-unit and divisible without the single-minded assumption and the single output restriction. They also generalize the results in Sun et al. (2019) to the case where the shippers' and carriers' bids are divisible, and the transaction costs are heterogeneous.

5. The case with homogeneous transaction costs

In many cases, the carriers gather in a logistics park. Then the transaction costs are only related to the locations of the shippers. In this case, a homogeneous transaction cost d_i is incurred when shipper i purchases transport service from any carrier. Then $d_{ij} = d_i, \forall j \in J$. In this case, the MMTR-MVCG double auction mechanism can be simplified further. In this section, we first describe a simplification of the MMTR-MVCG double auction mechanism. Then we design a new truthful double auction mechanism under the assumption that demand information is asymmetrical.

5.1. The MMTR-MVCG mechanism

In this subsection, we describe a simplification of the MMTR-MVCG mechanism in the case with homogeneous transaction costs. The maximum social welfare can be achieved by solving the linear programming ***LR2***(I, J), where d_{ij} is changed to d_i in ***LR1***(I, J).

When transaction costs are homogeneous, the lowest shipper bid price is not smaller than the highest carrier bid price in the efficient allocation. Then the shipper threshold price p^+ and the carrier threshold price p^- can be obtained as follows:

$$p^+ = \min\{\widehat{s}_i - d_i | \sum_{j\in J} q^*_{ij} > 0, i \in I\}, \tag{17}$$

$$p^- = \max\{\widehat{c}_j | \sum_{i\in I} q^*_{ij} > 0, j \in J\}, \tag{18}$$

where $q^*_{ij}(i \in I, j \in J)$ is the optimal solution of ***LR2***(I, J).

Note that shipper i pays the same transaction cost d_i while purchasing transportation service from any carrier, and the transaction costs do not require further adjustment as in Step 3 in the MMTR-MVCG mechanism. Then the shippers whose bid values are not higher than the shipper threshold price are removed directly.

After removing inefficient agents in the first stage, there are three scenarios regarding the relationship between the remaining shippers' total demand $Q(Q = \sum_{i\in\bar{I}} X_i)$ and the remaining carriers' total capacity $C(C = \sum_{j\in\bar{J}} Y_j)$: (1) $Q = C$, the total supply equals the total demand; (2) $Q > C$, the total demand exceeds the total supply; and (3) $Q < C$, the total supply exceeds the total demand. Then the rules that specify the shippers' payments and the carriers' revenue and the shipment allocation can be simplified as follows.

For scenario (1), the rule is as follows:

Rule 1: $\widetilde{I} = \bar{I}, \widetilde{J} = \bar{J}, \widetilde{P}_i = \sum_{j\in\bar{J}} q^*_{ij}(p^+ + d_i)$, and $\widetilde{P}_j = \sum_{i\in\bar{I}} q^*_{ij}(p^- + d_i)$.

For scenario (2), the rule is as follows:

Rule 2: All the remaining carriers trade, i.e., $\widetilde{J} = \bar{J}$, carrier j's revenue $\widetilde{P}_j = \sum_{i\in\bar{I}} q^*_{ij}(p^- + d_i)$. The final trading shipper set is $\widetilde{I} = \{i | \sum_{j\in\bar{J}} q^*_{ij} > 0, i \in \bar{I}\}$. Shipper i's payment $\widetilde{P}_i$ is calculated by (15).

For scenario (3), the rule is as follows:

Rule 3: All the remaining shippers trade, i.e., $\widetilde{I} = \bar{I}$, shipper i's payment $\widetilde{P}_i = \sum_{j\in\bar{J}} q^*_{ij}(p^+ + d_i)$. The final trading carriers set is $\widetilde{J} = \{j \big| \sum_{i\in\bar{I}} q^*_{ij} > 0, j \in \bar{J}\}$. Carrier j's revenue $\widetilde{P}_j$ is calculated by (16).

Now we describe the simplifying MMTR-MVCG mechanism for the case with homogeneous transaction costs in detail. In this case, Steps 2–8 in the MMTR-MVCG mechanism in Section 4 will be changed to Steps 2'–8'.

Step 2': Find the optimal solution q^*_{ij} of ***LR2***(I, J) and set the shipper threshold price p^+ and the carrier threshold price p^- by (17) and (18), respectively.

Step 3': Determine the remaining shipper set $\bar{I} = \{i | \widehat{s}_i - d_i > p^+, i \in I\}$ and the remaining carrier set $\bar{J} = \{j | \widehat{c}_j < p^-, j \in J\}$.

Step 4': Calculate the total remaining shippers' demand $Q(Q = \sum_{i\in\bar{I}} X_i)$ and the total remaining carriers' capacity $C(C = \sum_{j\in\bar{J}} Y_j)$.

Winner determination rule and payment rule

Steps 5'–7': Use Rule 1 for scenario (1), Rule 2 for scenario (2), and Rule 3 for scenario (3) to determine the trading agents and shippers' payments and carriers' revenue respectively.

Matching rule

Step 8': Trading shipper $i(i \in \widetilde{I})$ purchases q^*_{ij} units of transport service from trading carrier $j(j \in \widetilde{J})$.

Theorems 5–6 show that the simplifying MMTR-MVCG mechanism has incentive compatibility, individual rationality, budget balance, and asymptotic efficiency.

Theorem 5. *The simplifying MMTR-MVCG mechanism is incentive-compatible.*

Proof. Please see Appendix A.5.

Similar to Theorems 3–4, we can easily obtain Theorem 6.

Theorem 6. *The simplifying MMTR-MVCG mechanism is (ex post) individually rational, (ex post weakly) budget-balanced, and asymptotically efficient.*

5.2. Asymmetric demand information

In this subsection, we consider the case that the transaction costs are homogeneous and the demand information is private. The shippers will purchase multi-unit transport services. Suppose that their bids are dividable. Each carrier can only provide a single unit of transport service. This assumption is commonly used in the literature on double auction mechanisms, e.g., Babaioff and Walsh (2005), Shen (2006), Chu and Shen (2008), Cheng et al. (2016), and Sun et al. (2019). This assumption is consistent with the reality in China, since the majority of carriers are individual drivers and have a single unit of supply according to the IPO Prospectus of Full Truck Alliance (SEC, 2021). That is, shipper i may misrepresent demand quantity $\widehat{X}_i$ and $Y_j = 1$, $\forall j \in J$.

For the MMTR-MVCG mechanism, we can easily construct a counter-example that a shipper will lie and bid a smaller demand, which results in a lower shipper threshold price and a higher utility for him. Then the MMTR-MVCG mechanism cannot ensure incentive compatibility in this case. To circumvent this difficulty, we propose a new double auction mechanism for it (the DA-AQI mechanism).

Under the DA-AQI mechanism, we first introduce a virtual shipper i^* with X_I units of transport service and unlimited budgets. X_I is the largest demand quantity from a single shipper, which can be recorded by the platform through shippers' bids in cumulative historical auctions. Then we make a pricing decision for shippers, which helps remove some shippers from the auction. For shipper a, we find his threshold price p_a^+ according to (19), where $(q_{ij}^*)^{-a}$is the optimal solution of ***LR2***$(I\backslash\{a\}\cup\{i^*\}, J)$.

$$p_a^+ = \min\left\{\widehat{s}_i - d_i \middle| \left(q_{ij}^*\right)^{-a} > 0, i \in I\{a\}, j \in J\right\}, a \in I. \tag{19}$$

The remaining shipper set $\overline{I}$ can be determined as $\overline{I} = \{i|\widehat{s}_i - d_i > p_i^+, i \in I\}$.

Next, we determine the remaining carrier set $\overline{J}$. We set the carrier threshold price p^- as the lowest shipper threshold price, that is $p^- = \min\{p_i^+|i \in \overline{I}\}$. Then the remaining carrier set $\overline{J}$ can be determined as $\overline{J} = \{j|\widehat{c}_j < p^-, i \in I\}$.

After determining the remaining agent set $\overline{I}$ and $\overline{J}$, we adopt the TR-QD rule proposed by Liang et al. (2020) to match shipments and supplies, which records the time at which the agents submit their bids. Under this matching rule, agents who bid earlier have priority in the final allocation. Matching considering the bidding time can induce agents to bid price and quantity information truthfully, and quickly.

Now we state the procedure of the DA-AQI mechanism in detail:

Step 1: Shipper i submits a sealed bid $\{\widehat{s}_i, \widehat{X}_i\}$ and carrier j submits a sealed bid $\{\widehat{c}_j, \widehat{Y}_j\}$. The platform records each agent's bidding time.

Step 2: Use the perturbation technique to make the total bid price $\widehat{s}_i - d_i$ of shipper i and the bid price $\widehat{c}_j$ of carrier j different.

Step 3: Calculate each shipper i's $(i \in \overline{I})$ threshold price p_i^+ using (19) and determine the remaining shipper set, $\overline{I} = \{i|\widehat{s}_i - d_i > p_i^+, i \in I\}$.

Step 4: Set the carrier threshold price p^- as $\min\{p_i^+|i \in \overline{I}\}$ and determine the remaining carrier set, $\overline{J} = \{j|\widehat{c}_j < p^-, j \in J\}$.

Step 5: Use the remaining carriers' bids to meet the remaining shippers' bids according to the agents' bidding time.

Step 6: Trading shipper i gets the transportation services at the unit transaction price $p_i^+ + d_i$ and trading carrier j sells her transportation service at the unit transaction price, $p^- + d_i$ (the corresponding trading shipper i).

Now we prove that the DA-AQI mechanism has good properties.

Theorem 7. *The DA-AQI mechanism is incentive-compatible with respect to valuation and quantity.*

Proof. Please see Appendix A.6.

Theorem 8. *The DA-AQI mechanism is (ex post) individually rational and (ex post weakly) budget-balanced.*

Proof. Please see Appendix A.7.

Theorem 9. *Supposing that the valuation and cost distributions are bounded and the transaction costs are continuous, the DA-AQI mechanism is asymptotically efficient.*

Proof. Please see Appendix A.8.

Theorems 7–9 state that the DA-AQI mechanism can ensure incentive compatibility, individual rationality, budget balance, and asymptotical efficiency in the case with homogeneous transaction costs and asymmetric quantity information. They generalize the results in Babaioff and Walsh (2005) to the case with homogeneous transaction costs and the shippers' bids are multi-unit and divisible.

Table 3
An example of the DA-AQI mechanism.

Transportation market	
Shipper: ($\widehat{s}_i - d_i, \widehat{X}_i$,bidding time)	Carrier: ($\widehat{c}_j, \widehat{Y}_j$, bidding time)
(8, 1, 8:00 am)	(3, 1, 8:25 am)
(7, 2, 8:30 am)	(5, 1, 8:42 am)
(9, 2, 8:45 am)	(7, 1, 9:00 am)
(3, 1, 9:00 am)	(2, 1, 9:05 am)
(5, 3, 9:07 am)	(4, 1, 9:36 am)
(4, 2, 10: 00 am)	(6, 1, 9:45 am)
	(5, 1, 9:50 am)
	(4, 1, 10:00 am)

Now we give an example to illustrate how the DA-AQI mechanism works (see Table 3). In this example, the maximum social welfare is 22. We first set the demand X_I of the virtual shipper as 3. Then we make a pricing decision for shippers using (19). The threshold prices p_i^+ for shippers $i(i = 1, 2, ..., 6)$ are 7, 8, 7, 7, 7, and 7. Shippers 2, 4, 5, and 6 are removed from the trade and $\bar{I} = \{1,3\}$. Then we can find p^-=7 and the remaining carrier set is $\bar{J} = \{1,2,4,5,6,7,8\}$. Finally, the remaining shippers and the remaining carriers are matched in accordance with their bidding time. Shippers 1 and 2 purchase all of their demands at the price of 7 and carriers 1, 2, and 3 sell their transport service at the price of 7. The platform receives zero benefits and the mechanism efficiency is 72.73% (16/22). The above payment transfers do not include transaction costs.

6. Numerical studies

In this section, we study the impacts of transaction costs and demand information asymmetry on the performances on the allocation efficiency, and the equilibrium profits of the agents and the platform. We further compare the performance of our mechanisms with those of the current mechanism used by FOR-U Trucking (a famous online truck logistics platform in China) and the well-known MTR double auction mechanism.

6.1. Experimental parameters setting

Consider a transport lane from Beijing to Shanghai in China, with a total highway distance of 1,262 km, for which some shipments (to be transported from the shippers' locations to Shanghai) are auctioned. Some carriers who operate 9.6-meter trucks with a 13.5-tonne load compete for the shippers' demands. The shipments and the vehicles' depots are located in Beijing. The online freight platform, as the third-party auctioneer, operates the auctions periodically and clears the market through a specific auction mechanism. This platform makes decisions based on the bid prices and transaction costs related to the locations.

FOR-U Trucking is an online freight platform focusing on full truckload transport. The platform integrates a variety of transport resources, including its own fleets, franchised fleets, and individual drivers. At present, FOR-U Trucking matches the shipments and fleets by using the sequential reverse auction mechanism. A shipper places a demand order on the platform, and then the platform provides a real-time shipping quote. If the shipper accepts the quote, the platform will auction the demand to carriers using the first-price sealed reverse auction. The payoff of the platform comes from the difference between the quote by the platform and the money transferred to the carrier, for which the mechanism does not guarantee a budget balance. Under the MTR, MMTR-MVCG, and DA-AQI mechanisms, supplies and demands are matched in one shot using the double auction.

In a typical transaction, the carriers often offer a reference price for one fixed lane. Then they negotiate a transaction price with the shippers, which is usually higher than the reference price. Here we obtain the reference price ($/ton) from the official websites of some carriers around Beijing as their real valuations, which follow the continuous uniform distribution U_D[39,60]. FOR-U Trucking uses big data technology to predict shipping costs in real time within a 10% deviation from the market price (FOR-U Trucking, 2019). As a result, we set the market price ($) to fluctuate within 10% of the weighted average price of the carriers' transport costs. We derive the shippers' real valuations ($) from the market price within a 10% level of fluctuation. Under FOR-U Trucking mode, the carriers increase their quotations by 3.5−5% of their real valuations, which is the gross margin in the Chinese freight trucking industry (IResearch, 2019). The shippers need not report their valuations under FOR-U Trucking mode. Under our mechanisms, by Theorems 2, 5, and 7, the shippers and the carriers reveal their real valuations truthfully.

The carriers and shippers are in different locations, which results in different transaction costs. To facilitate computation, we use the costs of picking up the shipments as transaction costs, paid by the trading shippers. The costs are proportional to the trading volume, distance to pick-up, and transaction cost rates. To investigate the performance of the mechanisms in different trading environments, we consider Scenarios A and B corresponding to homogeneous transaction costs and heterogeneous transaction costs, respectively. Furthermore, the value range of the transaction cost rates increases from $0.015/ton-km to $0.075/ton-km in steps of $0.015/ton-km. We consider Scenario C for the case that the shippers will purchase multi-unit transport services and each carrier can only provide a single unit of transport service.

Scenario A: Transaction costs are heterogeneous. There are 20 shippers and 10 carriers distributed within Beijing city ($115.7°E - 117.4°E$, $39.4°N - 41.6°N$). We randomly generate their latitudes and longitudes. The transport demand quantities $X_i(i = 1, 2, ..., 20)$ follow the discrete uniform distribution U_D[1,3] and transport capacities $Y_j(j = 1, 2, ..., 10)$ follow the discrete uniform distribution

Table 4
The impact of transaction costs on mechanism efficiency in different scenarios.

Transaction cost rate			0.015	0.030	0.045	0.060	0.075
Scenario A	Maximum social welfare		1696.03	1499.05	1334.48	1176.09	1081.29
	Efficiency	FOR-U	72.34	58.33	41.77	25.47	14.41
		MMTR	**93.42**	**87.09**	**79.17**	**70.98**	**65.12**
Scenario B	Maximum social welfare		1366.01	1128.33	963.61	822.11	725.45
	Efficiency	FOR-U	78.43%	76.44%	68.18%	60.18%	52.86%
		MTR	95.21%	94.58%	93.87%	92.68%	91.74%
		MMTR	**98.07%**	**97.68%**	**97.09%**	**96.27%**	**95.20%**
		AQI	86.05%	84.82%	83.38%	80.82%	79.01%

Table 5
The impacts of transaction costs on trading results in different scenarios.

Transaction cost rate			0.015	0.030	0.045	0.060	0.075
Scenario A	Trading volume	FOR-U	14.16	9.34	5.79	3.14	1.61
		MMTR	**15.25**	**13.08**	**10.91**	**9.00**	**7.62**
	Trading shipper number	FOR-U	7.09	4.68	2.94	1.58	0.80
		MMTR	**7.92**	**6.86**	**5.84**	**4.79**	**4.09**
	Trading carrier number	FOR-U	3.14	1.86	0.99	0.41	0.17
		MMTR	**3.95**	**3.54**	**3.07**	**2.67**	**2.36**
Scenario B	Trading volume	FOR-U	**14.76**	10.44	7.17	5.28	3.93
		MTR	13.00	**11.36**	**9.94**	**8.69**	**7.74**
		MMTR	13.00	**11.36**	**9.94**	**8.69**	**7.74**
		AQI	12.43	10.82	9.45	8.19	7.24
	Trading shipper number	FOR-U	**7.34**	5.18	3.56	2.64	1.98
		MTR	6.55	**5.73**	**5.01**	**4.44**	**3.95**
		MMTR	6.55	**5.73**	**5.01**	**4.44**	**3.95**
		AQI	6.17	5.37	4.69	4.10	3.63
	Trading carrier number	FOR-U	**14.76**	10.44	7.17	5.28	3.93
		MTR	13.00	**11.36**	**9.94**	**8.69**	**7.74**
		MMTR	13.00	**11.36**	**9.94**	**8.69**	**7.74**
		AQI	12.43	10.82	9.45	8.19	7.24

$U_D[2,6]$.

Scenario B: Transaction costs are homogeneous. There are 20 shippers distributed within Beijing city ($115.7°E - 117.4°E$, $39.4°N - 41.6°N$) and 10 carriers gathered in Xinfadi, Beijing, at the latitude and longitude ($116.3550°E$, $39.8164°N$). We randomly generate shippers' latitudes and longitudes. The transport demand quantities $X_i (i = 1, 2, \ldots, 20)$ follow the discrete uniform distribution $U_D[1,3]$ and each carrier can only provide a single unit of transport service.

To investigate the impacts of demand information asymmetry on our mechanisms, we consider Scenario C similar to the market configuration under Scenario B but having the following difference.

Scenario C: The transaction cost rate is set to \$0.030/ton-km. The numbers of shippers and carriers increase from (10, 20) to (25, 35) in steps of (5, 5). We randomly select a shipper with 2-unit transport demand who will lie about his demand.

6.2. *Performance comparison of different mechanisms*

In this subsection, we study the impacts of transaction costs and demand information asymmetry on the performances (allocation efficiency, the equilibrium profits of the agents and the platform, trading volume, and trading agent number) of the two mechanisms. We further compare the performances of our mechanisms with those of the mechanism used by FOR-U Trucking and the MTR mechanism in three scenarios.

In the numerical studies, we have the following settings for each mechanism: (i) under the FOR-U Trucking mode, the carriers would add transaction costs to the bids after knowing the shippers' sites and the platform would incorporate the average transaction costs into the forecast quote for shippers. In addition, the shipping quote to the shipper by the platform is independent of the shipper's bid. So, we only consider the FOR-U mechanism in Scenarios A and B; (ii) the MTR mechanism cannot work under Scenario A. So, we only consider the MTR mechanism in Scenarios B and C; (iii) the MMTR-MVCG mechanism can be run in all Scenarios; and (iv) the DA-AQI mechanism can only be run in Scenarios B and C, where X_I is set to three. We repeat the auctions 1,000 times and report the average values.

We use FOR-U for the FOR-U Trucking mode and MMTR (AQI) for the MMTR-MVCG (DA-AQI) mechanism in Tables 4–7. The impacts of transaction costs on the mechanism efficiency in Scenarios A and B are illustrated in Table 4. "Maximum social welfare" represents the (average) maximal feasible social welfare in different scenarios and "Efficiency" (%) represents the (average) efficiency realized by the four mechanisms.

Table 4 clearly shows that the maximum social welfare and efficiencies of the four mechanisms decrease as the transaction cost rate

Table 6
The impacts of transaction costs on division of social welfare in different scenarios.

Transaction cost rate		0.015	0.030	0.045	0.060	0.075
Scenario A	FOR-U	**72.34**(893.01)	**58.86**(521.37)	**51.12**(289.19)	**46.91**(143.57)	40.59(66.36)
	MMTR	66.88(**1059.8**)	57.23(**752.20**)	49.63(**532.64**)	45.57(**391.48**)	**42.82**(**314.15**)
	FOR-U	27.66(341.53)	41.14(364.35)	48.88(276.57)	53.09(162.49)	**59.41**(97.14)
	MMTR	**33.12**(**524.73**)	**42.77**(**562.26**)	**50.37**(**540.55**)	**54.43**(**467.54**)	57.18(**419.55**)
Scenario B	FOR-U	87.73(941.20)	70.52(611.90)	59.74(400.14)	55.36(284.11)	50.61(205.34)
	MTR	89.40(**1162.8**)	89.21(**953.03**)	88.46(**800.62**)	87.57(**668.25**)	85.87(**573.44**)
	MMTR	84.22(1128.3)	83.74(923.26)	82.44(771.38)	80.90(641.07)	79.23(548.47)
	AQI	**98.4**1(1157.5)	**98.36**(942.51)	**98.40**(793.15)	**98.21**(655.84)	**97.96**(567.52)
	FOR-U	12.27(131.60)	**29.48**(**255.77**)	**40.26**(**269.62**)	**44.64**(**229.12**)	**49.39**(**200.38**)
	MTR	10.60(137.86)	10.79(115.26)	11.54(104.42)	12.43(94.82)	14.13(94.34)
	MMTR	**15.78**(**211.38**)	16.26(179.26)	17.56(164.26)	19.10(151.37)	20.77(143.76)
	AQI	1.59(18.66)	1.64(15.72)	1.60(12.93)	1.79(11.93)	2.04(11.80)

Table 7
The impacts of information asymmetry on the agents' utilities and platform's payoff.

Market size	MTR			MMTR			AQI		
	S	OA	P	S	OA	P	S	OA	P
(10,20)									
-1	11.49	472.69	98.10	11.49	445.88	149.82	8.02	461.83	9.12
0	9.26	467.73	102.50	9.26	441.18	157.42	8.02	465.52	8.61
+1	-84.99	470.41	99.29	-84.99	441.14	160.92	-84.99	469.88	9.03
(15,25)									
-1	9.06	719.65	104.01	9.06	686.72	165.17	6.88	705.64	14.60
0	8.14	720.43	106.66	8.14	689.26	165.70	6.88	706.28	14.58
+1	-82.90	713.37	112.25	-82.90	679.54	176.47	-82.90	708.67	12.33
(20,30)									
-1	8.90	919.44	116.33	8.90	887.60	181.69	7.33	912.90	15.02
0	8.13	918.91	115.97	8.13	887.26	184.08	7.33	912.98	13.82
+1	-85.43	921.30	116.67	-85.43	887.08	186.97	-85.43	916.54	14.35
(25,35)									
-1	7.60	1130.5	125.56	7.60	1099.4	192.61	6.43	1126.4	16.17
0	7.08	1136.7	122.77	7.08	1105.1	189.60	6.43	1131.9	16.86
+1	-83.57	1138.2	120.74	-83.57	1105.8	190.33	-83.57	1131.8	16.35

increases in both scenarios. The FOR-U mechanism's efficiency decreases fastest (decreasing from 72.34% to 14.41% when the transaction rate increases from \$0.015/ton-km to \$0.075/ton-km in Scenario A). The efficiencies of the other mechanisms decrease slowly (which means that the transaction costs do not have a large impact on the efficiency of the double auction mechanism). The reason is that the double auction mechanisms first collect bids from both shippers and carriers and then match supplies and demands with known transaction cost information, which can reduce the negative impacts of transaction costs. The efficiencies realized by the mechanisms sorted from the largest to the smallest are as follows: MMTR-MVCG, MTR, DA-AQI, and FOR-U. The MMTR-MVCG mechanism can help the online freight platform match the shippers' demands and the carriers' capacities more efficiently under both scenarios. Compared to the MMTR-MVCG, the DA-AQI gives up some efficiency to ensure incentive compatibility under asymmetric demand information.

Table 5 shows the impacts of transaction costs on the trading results in Scenarios A and B. "Trading volume" (%) represents the (average) quantity of transport services eventually exchanged under the four mechanisms. "Trading shipper number" and "Trading carrier number" represent the (average) number of shippers and carriers that can eventually trade under the four mechanisms, respectively.

From Table 5, the volume and agents traded under the four mechanisms decrease as the transaction cost rate increases in both scenarios. Those under the FOR-U mechanism decrease fastest but slowly under other mechanisms. In Scenarios A and B where the transaction cost rate is higher than \$0.015/ton-km, our mechanisms and the MTR mechanism can facilitate more agents to trade more deals than the FOR-U mechanism. In addition, although the MMTR-MVCG mechanism and the MTR mechanism realize the same trading volume and number of trading agents, the MMTR-MVCG mechanism yields more social welfare. This is because the matching rule under the MMTR-MVCG mechanism is different from that of the MTR mechanism, which makes the efficient resource allocation among the remaining agents.

The impacts of transaction costs on the division of social welfare in Scenarios A and B are shown in Table 6. "P" (%) represents the (average) percentage of welfare that the platform gets and "A" (%) represents the (average) percentage of agents' welfare. We also list the platform's payoff (\$) and agents' utility (\$) beside the percentage.

Table 6 demonstrates that the four mechanisms allocate more utility to the agents when the transaction cost is small in both scenarios. However, the FOR-U and MMTR-MVCG mechanisms tend to allocate more social welfare to the platform as the transaction

Table 8
Comparison of the characteristics of the four mechanisms.

			FOR-U	MTR	MMTR-MVCG	DA-AQI
Applied trading environment	Homogeneous transaction costs		√	√	√	√
	Heterogeneous transaction costs		√	×	√	×
	Information asymmetry on shipper's demand quantity		√	×	×	√
Performance	Heterogeneous transaction costs	Social welfare	Low	-	-	High
		Platform's payoff	Low	-	-	High
		Agents' utility	Low	-	-	High
		Trading volume	Small	-	-	Large
		Trading agent number	Small	-	-	Large
	Homogeneous transaction costs	Social welfare	Low	Medium	High	Medium
		Platform's payoff	High	Low	Medium	Low
		Agents' utility	Low	High	Medium	High
		Trading volume	Small	Large	Large	Large
		Trading agent number	Small	Large	Large	Large

cost rates increase in Scenario A. The reason is that the MMTR-MVCG mechanism sets a higher shipper threshold price and a lower carrier threshold price according to (9) and (10). Nevertheless, the agents' utility and platform's payoff under the MMTR-MVCG mechanism are larger than those under the FOR-U mechanism. In addition, as the transaction cost rates increase in Scenario B, our mechanisms allocate more utility to the agents while the FOR-U mechanism allocates more profits to the platform. Thus, our mechanisms can attract more potential shippers and carriers to participate in the auctions. Compared to the MTR mechanism, the MMTR-MVCG mechanism allocates more social welfare to the platform. Among the four mechanisms, the platform makes the least revenue under the DA-AQI mechanism, in which about 98% of the social welfare is allocated to the agents.

The impacts of demand information asymmetry on the agents' utilities and the platform's payoff are shown in Table 7. We use "(shipper number, carrier number)" to measure "market size". "0", "-1", and "+1" represent the selected shipper bidding truthfully, over-stating the demand for one unit and under-stating the demand for one unit, respectively. "S" ($) represents the selected shipper's (average) utility per trading unit and "OA" ($) represents the (average) utility of the agents except the selected shipper.

From Table 7, the shipper cannot get more benefit from lying about his demand under the DA-AQI mechanism, which is in line with Theorem 7. However, the shipper can benefit from under-reporting his demand but not from over-reporting his demand under the other two mechanisms. When the market size is very small, the platform may lose some profit under the MTR and MMTR-MVCG mechanisms if the shipper under-reports his demand. This is because a low shipper transaction price may be set. As the market size increases, the shipper's utility increment obtained by under-reporting demand under the MMTR-MVCG mechanism decreases. This means that shippers tend to bid truthfully under the MMTR-MVCG mechanism as the market size increases.

Table 8 summarizes the performance of the four mechanisms and demonstrates the advantages of our mechanisms. The MMTR-MVCG mechanism has good properties such as (*ex post*) individual rationality, (*ex post* weakly) budget balance, incentive compatibility, and asymptotical efficiency in the bilateral exchange environment with homogeneous or heterogeneous transaction costs. Moreover, the MMTR-MVCG mechanism is more efficient than the FOR-U and MTR mechanisms. When the transaction costs are heterogeneous, its performance (the profits of the agents and the platform, trading volume, and trading agent number) is better than that of the FOR-U mechanism. When the transaction costs are homogeneous, it allocates more welfare to the agents than the FOR-U mechanism and allocates more welfare to the platform than the MTR mechanism. Hence, our mechanism provides an easily implemented and efficient mechanism for online freight platforms in a transport market with transaction costs.

The DA-AQI mechanism relaxes the assumption that demand quantities are common knowledge and has the properties of (*ex post*) individual rationality, (*ex post* weakly) budget balance, incentive compatibility, and asymptotical efficiency in the bilateral exchange environment with homogeneous transaction costs. It can be applied to a transport market where the shippers' demand information is private. If the market is sufficiently large, the online freight platform can choose the MMTR-MVCG mechanism to clear the market because it yields more social welfare and allocates more profit to the platform.

7. The case with subsidy

To attract more suppliers and carriers to use their platforms, some online freight platforms provide subsidies for the agents using their platforms. In this section we consider the subsidy strategy and analyze the performance of the MMTR-MVCG mechanism. We consider a market similar to that in subsection 5.1.

Under the subsidy strategy, the platform announces that it will provide a subsidy $\alpha_I(\alpha_J)$ per trading unit to trading shippers (carriers) before the agents bid. In this case, shipper i's ($i \in I$) real valuation will be $s_i + \alpha_I$ and carrier j's ($j \in J$) real cost will be $c_j - \alpha_J$. Then, we have the following lemma, which shows the agents' equilibrium bidding strategies.

Lemma 1. *Under the subsidy strategy, shipper i's ($i \in I$) dominant strategy is to bid $\widehat{s}_i = s_i + \alpha_I$ and carrier j's ($j \in J$) dominant strategy is to bid $\widehat{c}_j = c_j - \alpha_J$.*

Now we discuss the impacts of the subsidy strategy on the performance of the MMTR-MVCG mechanism. Let $(\widetilde{I})^S$ ($(\widetilde{J})^S$) be the trading shipper (carrier) set and $(\widetilde{P}_i)^S$ ($(\widetilde{P}_j)^S$) be shipper i's transaction payment (carrier j's transaction revenue) under the MMTR-

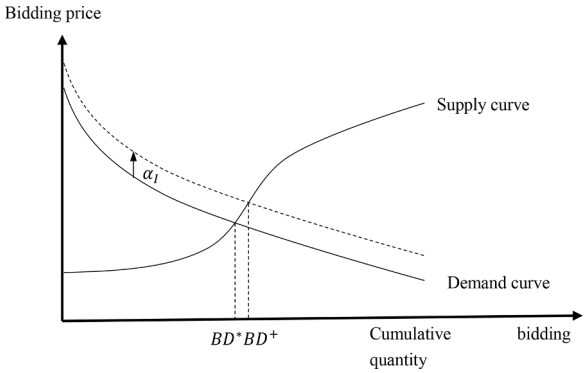


Fig. 4. Bilateral exchange market with the subsidy α_I

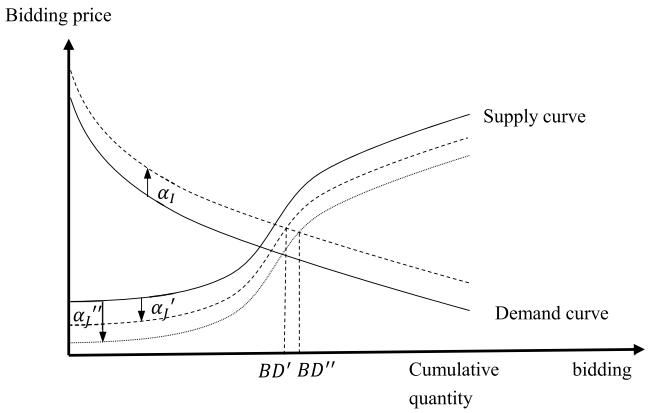


Fig. 5. Bilateral exchange market with a fixed α_I and α_J increasing.

MVCG mechanism with the subsidy strategy. If the platform provides a subsidy only for shippers, we have the following proposition.

Proposition 1. *If $\alpha_I > 0$ and $\alpha_J = 0$, then (i) $\widetilde{I} \subseteq (\widetilde{I})^S$ and $\widetilde{J} \subseteq (\widetilde{J})^S$; (ii) $0 \le (\widetilde{P}_i)^S - \widetilde{P}_i \le \alpha_I \sum_{j \in \widetilde{J}} q_{ij}^*$ for each shipper $i \in \widetilde{I}$ and $(\widetilde{P}_j)^S \ge \widetilde{P}_j$ for each carrier $j \in \widetilde{J}$.*

Proof. Please see Appendix A.9.

From Proposition 1, we know that there are more trading agents when the platform provides a subsidy for shippers. However, the profit increment of the trading shippers is less than the subsidy α_I. The reason is that the shippers' over-biddings drive the transaction price up. Although the platform does not subsidize the carriers directly, they can benefit from the subsidy for shippers because of more carrier trading and increasing transaction prices.

Similarly, if the platform provides a subsidy only for the carriers, we have the following proposition.

Proposition 2. *If $\alpha_J > 0$ and $\alpha_I = 0$, then (i) $\widetilde{I} \subseteq (\widetilde{I})^S$ and $\widetilde{J} \subseteq (\widetilde{J})^S$; (ii) $0 \le \widetilde{P}_j - (\widetilde{P}_j)^S \le \alpha_J \sum_{i \in \widetilde{I}} q_{ij}^*$ for each carrier $j \in \widetilde{J}$ and $(\widetilde{P}_i)^S \le \widetilde{P}_i$ for each shipper $i \in \widetilde{I}$.*

Proof. Please see Appendix A.9.

From Proposition 2, we know that there are more trading agents when the platform provides a subsidy for the carriers. In addition, the trading carriers obtain less additional utility than subsidy α_J. The reason is that the carriers' under-biddings drive the transaction price down. Although the platform does not subsidize shippers directly, they can benefit from the subsidy because more shippers trade and the transaction prices decrease.

If the platform provides subsidies for both the carriers and the shippers, we have the following proposition.

Proposition 3. *(1)* Given $\alpha_J \ge 0$, *when α_I increases, we have*

(i) $|(\widetilde{I})^S|$ *and* $\left|(\widetilde{J})^S\right|$ *are non-decreasing;*

(ii) $(\widetilde{P}_i)^S (i \in \widetilde{I})$ *and* $(\widetilde{P}_j)^S (j \in \widetilde{J})$ *are non-decreasing.*

(2) Given $\alpha_I \ge 0$, when α_J increases, we have

(i) $|(\widetilde{I})^S|$ *and* $\left|(\widetilde{J})^S\right|$ *are non-decreasing;*

(ii) $(\widetilde{P}_i)^S (i \in \widetilde{I})$ *and* $(\widetilde{P}_j)^S (j \in \widetilde{J})$ *are non-increasing.*

Proof. Please see Appendix A.10.

From Proposition 3, the higher the subsidies the platform provides, the more agents trade. In addition, the subsidy paid to the shippers can drive the transaction prices up and the subsidy paid to the carriers can drive the transaction prices down. Therefore, the platform can manipulate the market price by adjusting the subsidies α_I and α_J.

From the above analysis, the subsidy strategy can attract more potential shippers and carriers to participate in online platform trading. Because the MMTR-MVCG mechanism is asymptotically efficient, then the subsidy strategy can improve platform trading efficiency in the long run. Of course, the subsidy strategy may decrease the platform's profit and may result in a budget deficit.

8. Conclusion

In recent years, online freight platforms have developed rapidly and played an important role in the transport market. A critical problem faced by such platforms is how to induce the shippers and carriers to reveal their information truthfully and to match the shippers' demands and the carriers' supplies efficiently. Motivated by this problem, we propose in this paper two truthful double auction mechanisms for the online freight platform under the assumption that transaction costs are non-negative and the shippers' demand information is asymmetric. For the general case, we design a new mechanism, the MMTR-MVCG mechanism, and show that it is (*ex post*) individually rational, (*ex post* weakly) budget-balanced, incentive-compatible, and asymptotically efficient. For the special case with homogeneous transaction costs, we first simplify the MMTR-MVCG mechanism, which can reduce the computational burden. Furthermore, we consider asymmetric demand information and propose a new DA-AQI mechanism for the special case with homogeneous transaction costs. We show that it is (*ex post*) individually rational, (*ex post* weakly) budget balanced, incentive compatible, and asymptotically efficient. In addition, we generalize the MMTR-MVCG mechanism to include the subsidy strategy and show that the subsidy can attract more shippers and carriers and improve trading efficiency in the long run.

We conduct numerical studies to compare the performance of our mechanisms, the MTR mechanism, and the mechanism used by FOR-U Trucking. We obtain the following findings: (1) transaction costs decrease the efficiencies of the four mechanisms, and our mechanisms are less affected; (2) the MMTR-MVCG mechanism is more efficient than the other three mechanisms; (3) our mechanisms can yield greater social welfare, more trading agents, and larger trading volumes than the FOR-U Trucking mechanism; (4) our mechanisms tend to allocate more profits to the agents in the trading environment with homogeneous transaction costs, which can attract more shippers and carriers to participate in the online platform trading; and (5) the MMTR-MVCG mechanism is incentive-compatible with respect to quantity when the market is sufficiently large. This work provides easy implementation and efficient mechanisms (more efficient than the FOR-U mechanism, MTR mechanism) for the online freight platforms in a transport market with transaction costs and asymmetric demand information.

Of course, our work has some limitations. First, our mechanisms do not work in the transport market with heterogeneous transaction costs and asymmetric demand information. In this case, the MMTR-MVCG mechanism cannot ensure incentive compatibility and the DA-AQI mechanism may generate a budget deficit. So designing a mechanism with the desired properties for the market with heterogeneous transaction costs and asymmetric demand information remains an interesting and challenging topic for future research.

Second, to incentivize bidding demand truthfully, the DA-AQI mechanism has to give up some of the social welfare. It decreases the trading efficiency in a small transport market. Thus designing efficient mechanisms for the case with private demand information is another interesting topic.

Third, we do not consider collusion among the agents. The problem of collusion is prevalent in practice. Colluding bidders, also called cartel members, suppress bids or bid at non-competitive levels to designate the winners and transaction prices (Che et al., 2018). To design a truthful double auction mechanism, everyone who trades always pays (or sells at) the price proposed by someone else. Collusion among the agents can easily interfere with transaction prices. However, hidden bid-rigging is difficult to detect by platforms. Therefore, designing collusion-proof double auction mechanisms is a challenging topic.

In addition, the decision-making of the mechanisms is mainly based on the agents' bidding prices and transaction costs. In a real transport market, non-price attributes of the transport service can also affect the trading relationships between shippers and carriers, such as the type of trucks, the nature of the goods, time windows, etc. Another interesting research topic is to incorporate these non-price attributes into the design of the double auction mechanism

CRediT authorship contribution statement

Jiantao Guo: Conceptualization, Methodology, Writing – original draft. **Juliang Zhang:** Supervision. **T.C.E. Cheng:** Writing – review & editing. **Shouting Zhao:** Investigation.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under grant numbers 72171016 and 71661167009; NSFC/RGC joint support scheme under grant number 3-RAA7; and the Beijing Logistics Informatics Research Base.

Cheng was also supported in part by The Hong Kong Polytechnic University under the Fung Yiu King-Wing Hang Bank Endowed Professorship in Business Administration.

Appendix

A.1. Proof of Theorem 1

To prove this theorem, we first introduce the concept of *totally unimodular matrix* and two lemmas.

Definition 1. *(Camion, 1965). Let A be a real matrix. We call A a totally unimodular matrix if every square submatrix of A has the determinant equal to 0, 1, or -1.*

Lemma 2. *(Garfinkel and Nemhauser, 1972). A matrix* $A = (a_{ij})_{m \times n}$ *with* $a_{ij} = 0, 1$ *or -1 is totally unimodular matrix if no more than two non-zero elements appear in each column, and the rows can be partitioned into two subsets* I_1 *and* I_2*, such that*

(1) *if a column contains two non-zero elements with the same sign, the elements are in different subsets, or*
(2) *if a column contains two non-zero elements of opposite sign, both elements are in the same subset.*

Lemma 3. *(Georgiou and Floudas, 1989). For a linear program* ***LP***

$$\begin{aligned} \boldsymbol{LP} \quad & \max(or\min) z = c^T x \\ & s.t. \quad Ax = b \\ & \quad\; x \geq 0 \end{aligned}$$

If matrixA is a totally unimodular matrix and b is an integer vector, then the fundamental solution $\widehat{x} = (\widehat{x}_1, \widehat{x}_2, \widehat{x}_3, \ldots, \widehat{x}_m)^T$ *of* ***LP*** *is an integer vector.*

Now we prove Theorem 1.

Note that the coefficient matrix $\boldsymbol{A}$ of linear program ***LR1***(I, J) is as follows:

$$\boldsymbol{A} = \begin{array}{c} \\ 1 \\ 2 \\ \vdots \\ m \\ m+1 \\ m+2 \\ \vdots \\ m+n \end{array} \begin{array}{c} \begin{array}{cccccccccccccc} q_{11} & q_{12} & \cdots & q_{1n} & q_{21} & q_{22} & \cdots & q_{2n} & \cdots & q_{m1} & q_{m2} & \cdots & q_{mn} \end{array} \\ \begin{pmatrix} 1 & 1 & \cdots & 1 & 0 & 0 & \cdots & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 1 & 1 & \cdots & 1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & \cdots & 1 & 1 & \cdots & 1 \\ 1 & 0 & \cdots & 0 & 1 & 0 & \cdots & 0 & \cdots & 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & 0 & 1 & \cdots & 0 & \cdots & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & 0 & 0 & \cdots & 1 & \cdots & 0 & 0 & \cdots & 1 \end{pmatrix} \end{array}.$$

Then matrix $\boldsymbol{A}$ is a totally unimodular matrix from Lemma 2. Furthermore, we can write the constraints of ***LR1***(I, J) in the following form

$$\begin{cases} (\boldsymbol{A}, \boldsymbol{I}) \begin{pmatrix} x \\ y \end{pmatrix} = b, \\ x \geq 0, y \geq 0, \end{cases}$$

where $\boldsymbol{I}$ is a unit matrix and $\boldsymbol{A}$ is a totally unimodular matrix. So $(\boldsymbol{A}, \boldsymbol{I})$ is also a totally unimodular matrix. $b = (X_1, X_2, \ldots, X_m, Y_1, Y_2, \ldots, Y_n)^T$ consists of the shippers' transport demands and the carriers' transport capacity, and is an integer vector. From Lemma 3, the optimal solution $(q_{ij}^*, i \in I, j \in J)$ to linear program ***LR1***(I, J) is an integer vector.

A.2. Proof of Theorem 2

Suppose that carrier $i_0 (i_0 \in I)$ with real value s_{i_0} submits a bid $\{\widehat{s}_{i_0}, X_{i_0}\}$ and the other agents bid truthfully. We use $\widehat{h}_{ij}$ to denote shipper i's total bid price, i.e., $\widehat{h}_{ij} = \widehat{s}_i - d_{ij}$. If shipper i_0 does not bid truthfully, there are two cases: (1) $\widehat{s}_{i_0} > s_{i_0}$, and (2) $\widehat{s}_{i_0} < s_{i_0}$.

Case (1), $\widehat{s}_{i_0} > s_{i_0}$.

Under the MMTR-MVCG mechanism, the shipper threshold price p^+ and the carrier threshold price p^- are set independently of the

remaining agents' bid prices. The shippers with the total bid prices less than or equal to p^+ and the carriers with bid prices greater than or equal to p^- are removed. The transaction prices are determined by the modified VCG mechanism for the remaining agents, which are higher than or equal to p^+ (lower than or equal to p^-).

For shipper i_0, let $H_{i_0} = \{\widehat{h}_{i_0 1}, \widehat{h}_{i_0 2}, ..., \widehat{h}_{i_0 n}\} = \{\widehat{s}_{i_0} - d_{i_0 1}, \widehat{s}_{i_0} - d_{i_0 1}, ..., \widehat{s}_{i_0} - d_{i_0 n}\}$. Then, H_{i_0} can be divided into two subsets $H^1_{i_0}$ and $H^2_{i_0}$ as follows:

$$H^1_{i_0} = \left\{\widehat{h}_{i_0 j} \middle| \widehat{h}_{i_0 j} > p^+, j \in J\right\}, \text{ and} H^2_{i_0} = \left\{\widehat{h}_{i_0 j} \middle| \widehat{h}_{i_0 j} \le p^+, j \in J\right\}. \tag{A.1}$$

On the one hand, shipper i_0 has a chance to trade with carrier j with $\widehat{h}_{i_0 j} > p^+$ in the final allocation, and his transaction price of trading with carrier j is determined by the modified VCG mechanism, which is well known to be incentive compatible (Groves, 1973; Krishna, 2002). Therefore, shipper i_0 will not manipulate his bid price.

On the other hand, if the trading relationship between shipper i_0 and carrier j_0 is removed in the first allocation, he cannot trade with carrier j_0 with $\widehat{h}_{i_0 j_0} \le p^+$. If he bids $\widehat{s}_{i_0}$ that makes $\widehat{h}_{i_0 j_0}$ higher than p^+, she can trade with carrier j_0 at the price of p^+ or higher. Then, she gets the utility $h_{i_0 j_0} - p^+$ per trading volume, which is negative. The bid price $\widehat{s}_{i_0}$ also makes $\widehat{h}_{i_0 j}(\widehat{h}_{i_0 j} \in H^1_{i_0})$ higher. Then shipper i_0 cannot get more utility by trading with carrier j with $\widehat{h}_{i_0 j} > p^+$ according to the modified VCG mechanism. In summary, shipper i_0's utility remains unchanged or even decreases if he bids higher than s_{i_0}. Therefore, he has no incentive to manipulate her bid price.

Case (2), $\widehat{s}_{i_0} < s_{i_0}$.

If shipper i_0 bids truthfully, H_{i_0}, the set of his bid price minus transaction costs for trading with each carrier j can be divided into two subsets $H^1_{i_0}$ and $H^2_{i_0}$ according to formulas (A.1). If he bids $\widehat{s}_{i_0}$ lower than s_{i_0}, then $\widehat{h}_{i_0 j} < h_{i_0 j}$ for some carriers. Some elements in $H^1_{i_0}$ will be in $H^2_{i_0}$. Shipper i_0 will not trade with carrier j such that $\widehat{h}_{i_0 j} \in H^2_{i_0}$ because the trading relationship between shipper i_0 and carrier j with $\widehat{h}_{i_0 j} \le p^+$ is removed in the first allocation. Therefore, if shipper i_0 bids $\widehat{s}_{i_0} < s_{i_0}$, his trade chance will be reduced. Moreover, the lower bid price $\widehat{s}_{i_0}$ will also decrease $h_{i_0 j} \in H^1_{i_0}$ to $\widehat{h}_{i_0 j}$. Then he cannot get more utility for trading with carrier j such that $\widehat{h}_{i_0 j} \in H^1_{i_0}$ because the transaction price is set by the modified VCG mechanism, which is incentive compatible.

Summarizing the above two cases, we know that the shippers will bid truthfully under the MMTR-MVCG mechanism.

By similar arguments, we can show that the carriers will also bid their real costs. Then the MMTR-MVCG mechanism is incentive compatible.

A.3. Proof of Theorem 3

Under the MMTR-MVCG mechanism, the shippers whose bid prices are less than or equal to the shipper threshold price p^+ and the carriers whose bid prices are greater than or equal to the carrier threshold price p^- are removed after the first stage. The final transaction prices are determined by the modified VCG mechanism, i.e., the utilities of the trading agents are determined by their contributions. Note that the trading agents' contributions are non-negative, so their utilities are non-negative. For trading shipper $i(i \in \widetilde{I})$, his total payment is $\widetilde{P}_i = p_i^{VCG} \le \sum_{j \in J} \widetilde{q}^*_{ij} \widehat{s}_i$. For trading carrier $j(j \in \widetilde{J})$, her total payment is $\widetilde{P}_j = p_j^{VCG} \ge \sum_{i \in I} \widetilde{q}^*_{ij}(\widehat{c}_j + d_{ij})$. From Theorem 2, the agents bid truthfully, i.e., $\widehat{s}_i = s_i$ and $\widehat{c}_j = c_j$. Then the trading agents get non-negative utilities. Thus, the MMTR-MVCG mechanism is individually rational.

From (9) and (10), the shipper threshold price p^+ is larger than or equal to the carrier threshold price p^-. According to the modified VCG mechanism, we replace the bid price of each remaining shipper (carrier) with the threshold price $p^+(p^-)$ after the first stage. For trading shipper $i(i \in \widetilde{I})$, we have $\widetilde{P}_i = p_i^{VCG} \ge \sum_{j \in J} \widetilde{q}^*_{ij}(p^+ + d_{ij})$. For trading carrier $j(j \in \widetilde{J})$, we have $\widetilde{P}_j = p_j^{VCG} \le \sum_{i \in I} \widetilde{q}^*_{ij}(p^- + d_{ij})$. Then,

$$\sum_{i \in \widetilde{I}} \widetilde{P}_i \ge \sum_{i \in \widetilde{I}} \sum_{j \in \widetilde{J}} \widetilde{q}^*_{ij}(p^+ + d_{ij}) \ge \sum_{j \in \widetilde{J}} \sum_{i \in \widetilde{I}} \widetilde{q}^*_{ij}(p^- + d_{ij}) \ge \sum_{j \in \widetilde{J}} \widetilde{P}_j.$$

The shippers' total payment is not less than the carriers' total revenue. The MMTR-MVCG mechanism is budget-balanced.

A.4. Proof of Theorem 4

In this section we prove the asymptotic efficiency of the MMTR-MVCG mechanism.

Suppose that there are m shippers and n carriers. Shipper i's demand $X_i(i = 1, 2..., m)$ and carrier j's capacity $Y_j(j = 1, 2..., n)$ are bounded, i.e., there are X' and Y' such that $X_i \le X'$ and $Y_j \le Y'$. Assume that shipper i's valuation follows $F(s_i)_{i \in I}$ with the support $[\underline{s}, \overline{s}]$ and carrier j's cost follows $G(c_j)_{j \in J}$ with the support $[\underline{c}, \overline{c}]$. The transaction cost d_{ij} is proportional to the distance between shipper i and carrier j, who are independently distributed according to some continuous distribution U on some compact domain H. The distance function is continuous with respect to the agents' locations.

To prove this theorem, we first prove the following lemma.

Lemma 4. *If $d_{ij} = d_i$, the MMTR-MVCG mechanism is asymptotically efficient.*

From the assumption, we have that the transaction cost d_i is proportional to the distance between shipper i and the logistics park

where the carriers are located. Let $a = \min\{s_i - d_i | i \in I\}$, $b = \max\{s_i - d_i | i \in I\}$. Without loss of generality, we assume that $b > \underline{c}$. Obviously, the MMTR-MVCG mechanism can realize the maximum social welfare among the remaining shippers and carriers. The efficiency loss is incurred by the removed agents. Let Δ be the welfare loss and E be the mechanism efficiency, so

$$E = \frac{\max V(\bar{I}, \bar{J})}{\max V(I, J)} = \frac{\max V(I, J) - \Delta}{\max V(I, J)}.$$

Let q_{ij}^* be the optimal solution of ***LR2***(I, J). Then we have the efficient shipper set $I^* = \{i \Big| q_{ij}^* > 0, i \in I\}$ and the efficient carrier set $J^* = \{j \Big| q_{ij}^* > 0. j \in J\}$, who are in the efficient allocation. Every agent's bid price is different according to the perturbation technique. So, only one shipper's total bid price is equal to the shipper threshold price p^+ and one carrier's bid price is equal to the carrier threshold price p^-. Let i_0 and j_0 be these two agents and then we have $\bar{I} = I^* \ i_0$ and $\bar{J} = J^* \ j_0$. Under the MMTR-MVCG mechanism, we have the trading shipper set $\widetilde{I} = \{i \Big| q_{ij}^* > 0, i \in \bar{I}, j \in \bar{J}\}$ and the trading carrier set $\widetilde{J} = \{j \Big| q_{ij}^* > 0, i \in \bar{I}. j \in \bar{J}\}$. The difference between the trading volume under the efficient allocation and the MMTR-MVCG mechanism is

$$\sum_{i \in I^*} \sum_{j \in J^*} q_{ij}^* - \sum_{i \in \widetilde{I}} \sum_{j \in \widetilde{J}} q_{ij}^* = \max\left\{ \sum_{i \in I^*} q_{ij_0}^*, \sum_{j \in J*} q_{i_0 j}^* \right\}.$$

Note that there are $\sum_{i \in I^*} q_{ij_0}^* \le Y_{j_0}$ and $\sum_{j \in J*} q_{i_0 j}^* \le X_{i_0}$ according to the constraints (6) and (7). Because X_i and Y_j are bounded, $\max\left\{ \sum_{i \in I^*} q_{ij_0}^*, \sum_{j \in J*} q_{i_0 j}^* \right\}$ are also bounded. Furthermore, a, b, $\underline{c}$, and $\bar{c}$ are bounded, so the efficiency loss Δ is bounded. The maximum welfare of the market will approach infinity as the number of agents approaches infinity, i.e.,

$$\lim_{m \to \infty, n \to \infty} \max V(I, J) = \infty.$$

Then,

$$\lim_{m \to \infty, n \to \infty} E = 1 - \lim_{m \to \infty, n \to \infty} \frac{\Delta}{\max V(I, J)} = 100\%.$$

Therefore, the MMTR-MVCG mechanism in this special case with homogeneous transaction costs is asymptotically efficient.

Now, we prove Theorem 4.

Suppose that shipper i_0's bid price minus his transaction cost is lower than all carriers' bid prices other than carrier j_0. She will trade with carrier j_0 in the first allocation. This means that the transaction costs between shipper i_0 and the carriers but carrier j_0 are much larger than the transaction cost between shipper i_0 and carrier j_0. Since the transaction cost d_{ij} is proportional to the distance between shipper i and shipper j, shipper i_0 and carrier j_0 are located closely and far from the other agents. Note that the agents are distributed on a compact domain H, so there exist finite partitions $A_1, A_2, \ldots, A_k$ of H (where a partition A_l has a radius ε_l). When the number of agents approaches infinity, the agents in a partition will trade with one another because of their lower transaction costs, and the trading volume in each partition is sufficiently large. If we set the radius ε_l such that $\inf\{\widehat{s}_i - \alpha\varepsilon_l | \sum_{j \in J} q_{ij}^* > 0, i \in I_{A_l}\} \ge \sup\{\widehat{c}_j | \sum_{i \in I} q_{ij}^* > 0, j \in J_{A_l}\}$ (where α is the cost-distance coefficient) in the first allocation of each partition A_l, we can show that the MMTR-MVCG mechanism is asymptotically efficient within each partition by similar arguments in the proof of Lemma 4. Therefore, the MMTR-MVCG mechanism is asymptotically efficient on the domain H.

A.5. Proof of Theorem 5

In the case with the homogeneous transaction costs, the shipper threshold price p^+ only depends on the shipper with the lowest bid price in the first allocation, and the carrier threshold price p^- only depends on the carrier with the highest bid price in the first allocation under the MMTR-MVCG mechanism. Then they are irrelevant to the remaining agents. The shippers whose total bid prices are less than or equal to p^+ and the carriers whose bid costs are greater than or equal to p^- are removed. Then shipper i's transaction payment $\widetilde{P}_i$ is greater than or equal to $\sum_{j \in \widetilde{J}} \bar{q}_{ij}^* p^+$ and carrier j's transaction revenue $\widetilde{P}_j$ is less than or equal to $\sum_{i \in \widetilde{I}} \bar{q}_{ij}^* p^-$. The agents removed in the first stage will get negative utilities if they finally trade by manipulating their bid prices, so they will reveal their valuations truthfully.

For the remaining agents, when the total supply and the total demand are equal, the transaction prices are determined by p^+ and p^-. The shippers (carriers) cannot get a lower (higher) transaction price by manipulating the bid prices. When the remaining total supply and the total remaining demand are not equal, the agents' transaction prices are determined by the modified VCG mechanism, which is an incentive compatible mechanism (Groves, 1973 Krishna, 2002;). Therefore, the shippers (carriers) do not manipulate their bid prices (costs) to get lower (higher) utilities.

A.6. Proof of Theorem 7

We first show that the agents will truthfully reveal their information under the DA-AQI mechanism.

We first consider the biding behaviors of shippers. In the DA-AQI mechanism, whether shipper i' can remain in $\bar{I}$ depends on $p_{i'}^{+}$, which is dependent on the agent set $I\backslash\{i'\}\cup\{i^*\}\cup J$ but independent of his bid. If shipper i' who is removed from the trade bids higher such that he remains in $\bar{I}$, he will get a negative utility ($p_{i'}^{+} > s_{i'}$) and she also has no incentive to bid falsely. We next show that all remaining shippers can purchase all of their demands. Let $(q_{ij}^{*})^{i*}$ be the optimal solution of ***LR2***$(I\cup\{i^*\}, J)$. Note that $\min\{\widehat{s}_i - d_i \mid (q_{ij}^{*})^{i*} > 0, i \in I, j \in J\} \geq \min\{p_i^{+} \mid i \in \bar{I}\} = p^{-}$. Then we have that $\sum_{j\in\bar{J}}\widehat{Y}_j \geq X_I + \sum_{i\in\bar{I},j\in J}(q_{ij}^{*})^{i*}$. If $\sum_{j\in J}(q_{ij}^{*})^{i*} = \widehat{X}_i$ for each remaining shipper $i \in \bar{I}$, $\sum_{j\in\bar{J}}\widehat{Y}_j > \sum_{i\in\bar{I},j\in J}(q_{ij}^{*})^{i*} = \sum_{i\in\bar{I}}\widehat{X}_i$. If there is a remaining shipper $i' \in \bar{I}$ with $\sum_{j\in J}(q_{i'j}^{*})^{i*} < \widehat{X}_{i'}$, $\sum_{j\in\bar{J}}\widehat{Y}_j \geq \sum_{i\in\bar{I}}\widehat{X}_i$ because $X_I \geq \widehat{X}_{i'}$. Then we can conclude that all remaining shippers will finally purchase all of their demands. Furthermore, transaction price of each trading shipper i' is $p_{i'}^{+}$, which is dependent on the agent set $I\backslash\{i'\}\cup\{i^*\}\cup J$. Therefore shippers will truthfully reveal their demand quantity information.

We next consider the biding behaviors of carriers. For carrier j' who is removed from the trade, if she bids lower such that she trades, she will get a negative utility ($p^{-} < c_{j'}$) and she also has no incentive to bid falsely. For remaining carrier j' ($j' \in \bar{J}$), whether she can trade finally is dependent on her bidding time. Furthermore, if remaining carrier j' finally trade, her transaction price is p^{-}. If she bids lower, p^{-} will keep unchanged. If she bids higher such that p^{-} is raised, she will be removed from the trade. Therefore, she also has no incentive to bid falsely.

Summarizing the above, we prove Theorem 7.

A.7. Proof of Theorem 8

Under the DA-AQI mechanism, the shippers whose bid prices are less than or equal to his shipper threshold price p_i^{+} and the carriers whose bid prices are greater than or equal to the carrier threshold price p^{-} are removed. The final transaction prices for trading shipper i is p_i^{+} and the trading carriers sell their services at the price p^{-}. All the trading agents will get non-negative utilities, so the DA-AQI mechanism is individually rational.

On the one hand, the transaction price of each remaining shipper $i(i \in \bar{I})$ is his threshold price p_i^{+}. On the other hand, all the remaining carriers' transaction prices are equal to the carrier threshold price p^{-}. Note that $p^{-} = \min\{p_i^{+} \mid i \in \bar{I}\}$, and the DA-AQI mechanism is budget-balanced.

A.8. Proof of Theorem 9

Suppose that there are m shippers and n carriers. Shipper i's demand $X_i(i = 1, 2..., m)$ and carrier j's capacity $Y_j(j = 1, 2..., n)$ are bounded. Assume that shipper i's valuation follows $F(s_i)_{i\in I}$ with support $[\underline{s}, \overline{s}]$ and carrier j's cost follows $G(c_j)_{j\in J}$ with support $[\underline{c}, \overline{c}]$. The transaction cost d_i is proportional to the distance between shipper i and the logistics park where the carriers are located. The distance function is continuous with respect to the shippers' locations. Let $a = \min\{s_i - d_i \mid i \in I\}$, $b = \max\{s_i - d_i \mid i \in I\}$. Without loss of generality, we assume that $b > \underline{c}$.

According to the DA-AQI mechanism, there is at most one efficient shipper removed from the trade. Let Δ be the welfare loss. We have that $\Delta \leq X_I \times (b - \underline{c})$. Because X_I, b, and $\underline{c}$ are bounded, the efficiency loss Δ is also bounded. Let E be the mechanism efficiency. It will approach 100% as the number of agents approaches infinity, i.e.,

$$\lim_{m\to\infty,n\to\infty} E = 1 - \lim_{m\to\infty,n\to\infty} \frac{\Delta}{\max V(I,J)} = 100\%.$$

Therefore, the DA-AQI mechanism is asymptotically efficient.

A.9. Proof of Propositions 1 and 2

We first consider the case with the subsidy α_I only for shippers. In this case, the demand curve $D(BQ)$ is constructed by sorting $s_i - d_i$ for all shippers $i \in I$ from largest to smallest, where BQ is the cumulative bidding quantity. The supply curve $S(BQ)$ is constructed by sorting $c_j(j \in J)$ from smallest to largest.

Let BQ^* be the equilibrium quantity, which can be found by $BQ* = \underset{BQ}{\operatorname{argmax}}\{BQ \mid D(BQ) \geq S(BQ)\}$. From Lemma 1, after the platform announces the subsidy strategy, the demand curve will be $D(BQ) + \alpha_I$. Let BQ^{+} be the equilibrium quantity with the subsidy α_I. Then $BQ^{+} = \underset{BQ}{\operatorname{argmax}}\{BQ \mid D(BQ) + \alpha_I \geq S(BQ)\}$. Let $(p^{+})^S((p^{-})^S)$ and $(\bar{I})^S((\bar{J})^S)$ be the shipper (carrier) threshold price and the remaining shipper (carrier) set with the subsidy α_I.

From Fig. 4, the subsidy α_I essentially shifts the demand curve $D(BQ)$ upwards by α_I. Then the new equilibrium price will be higher, that is $S(BQ^{+}) \geq S(BQ^*)$. Because $(p^{-})^S = S(BQ^{+})$, $(p^{-})^S$ will be higher than p^{-} and more carriers will be remained in $(\bar{J})^S$. With the higher equilibrium price, we have that $D(BQ^{+}) + \alpha_I \geq D(BQ^*)$. Then the new shipper threshold price will be higher because $(p^{+})^S = D$

$(BQ^+) + \alpha_I \geq D(BQ^*) = p^+$. Because $dS(BQ)/dBQ \geq 0$ and $dS(BQ)/dBQ$ is bounded, we have that $(p^+)^S - p^+ \leq \alpha_I$. More shippers will be remained in $|(\bar{I})^S|$. The MMTR-MVCG mechanism makes the efficient allocation among the remaining shippers and remaining carriers. With non-decreasing $|(\bar{I})^S|$ and $(\bar{J})^S$, we have that $\widetilde{I} \subseteq (\bar{I})^S$ and $\widetilde{J} \subseteq (\bar{J})^S$.

Next, we consider the case that $Q > C$ and show how the transaction payment and revenue change. According to the payment rule of the MMTR-MVCG mechanism, carrier j's ($j \in \widetilde{J}$) revenue with the subsidy α_I is $(\widetilde{P}_i)^S = \sum_{i\in\widetilde{I}} q^*_{ij}((p^-)^S + d_i)$, which is no lower than $\widetilde{P}_i = \sum_{i\in\widetilde{I}} q^*_{ij}(p^- + d_i)$ because $(p^-)^S \geq p^-$. Shipper i's ($i \in \widetilde{I}$) payment is $(\widetilde{P}_i)^S = \hat{s}_i \sum_{j\in\widetilde{J}} q^*_{ij} - V(\bar{I},\bar{J}\big|\hat{c}_j = (p^+)^S) + V(\bar{I},\bar{J}\big|\hat{c}_j = (p^+)^S)^{-i}$, which is no lower $\widetilde{P}_i$ because $(p^+)^S \geq p^+$. Furthermore, we have that $(\widetilde{P}_i)^S - \widetilde{P}_i \leq \alpha_I \sum_{j\in\widetilde{J}} q^*_{ij}$ for shipper i because $(p^+)^S - p^+ \leq \alpha_I$. In the other two cases, we can also show the same proposition. Thus, we have that $0 \leq (\widetilde{P}_i)^S - \widetilde{P}_i \leq \alpha_I \sum_{j\in\widetilde{J}} q^*_{ij}$ for each shipper $i \in \widetilde{I}$ and $(\widetilde{P}_j)^S \geq \widetilde{P}_j$ for each carrier $j \in \widetilde{J}$.

By similar arguments, we can prove Proposition 2.

A.10. Proof of Proposition 3

We first consider the case with a fixed α_I and α_J increasing. Let α_J' and α_J'' be the subsidies paid to the carriers ($\alpha_J'' > \alpha_J'$). Let $(\widetilde{I})^{S'}((\widetilde{J})^{S'})$ and $(\widetilde{I})^{S''}((\widetilde{J})^{S''})$, $(p^-)^{S'}$ and $(p^-)^{S''}$, $(\widetilde{P}_j)^{S'}$ and $(\widetilde{P}_j)^{S''}$ be the trading shipper (carrier) sets, carrier threshold prices, transaction revenues for carrier j with the subsidies α_J' and α_J'', respectively. Let BQ' and BQ'' be the equilibrium quantities with the subsidies α_J' and α_J'', respectively, where $BQ' = \underset{BQ}{\operatorname{argmax}}\{BQ|D(BQ) + \alpha_I \geq S(BQ) - \alpha_J'\}$ and $BQ'' = \underset{BQ}{\operatorname{argmax}}\{BQ|D(BQ) + \alpha_I \geq S(BQ) - \alpha_J''\}$.

From Fig 5, the subsidy α_J'' essentially shifts the demand curve with α_J' downwards by $\alpha_J'' - \alpha_J'$. Then the new equilibrium price will be lower. By similar arguments of Proposition 1, we have that $(\widetilde{I})^{S'} \subseteq (\widetilde{I})^{S''}$ and $(\widetilde{J})^{S'} \subseteq (\widetilde{J})^{S''}$. Therefore, $|(\widetilde{I})^S|$ and $\left|(\widetilde{J})^S\right|$ are non-decreasing when α_J increases.

In addition, because $dD(BQ)/dBQ \leq 0$ and $dD(BQ)/dBQ$ is bounded, $S(BQ'') - \alpha_J'' \leq S(BQ') - \alpha_J'$. Then we have that $(p^-)^{S''} \leq (p^-)^{S'}$ and $(p^+)^{S''} \leq (p^+)^{S'}$. According to the payment rule of the MMTR-MVCG mechanism, we have that $(\widetilde{P}_j)^{S''} \leq (\widetilde{P}_j)^{S'}$ for each carrier $j \in \widetilde{J}$ and $(\widetilde{P}_i)^{S''} \leq (\widetilde{P}_i)^{S'}$ for each shipper $i \in \widetilde{I}$. Therefore, when α_J increases, $(\widetilde{P}_i)^S (i \in \widetilde{I})$ and $(\widetilde{P}_j)^S (j \in \widetilde{J})$ are no-increasing.

By similar arguments, we can prove Proposition 3. (2) Eqn (A1).

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